

PMATH 450

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1 Riemann Integration

Definition 1.1. We say $\mathcal{P} = \{a_0, a_1, \dots, a_n\}$ is a partition of $[a, b]$ iff

$$a = a_0 < a_1 < \dots < a_n = b.$$

Definition 1.2. $f : [a, b] \rightarrow \mathbb{R}$ bounded. For a partition $\mathcal{P} = \{a_0, \dots, a_n\}$, we define the lower and upper sums by

$$L(f, \mathcal{P}) = \sum_{i=1}^n m_i(a_i - a_{i-1})$$

$$U(f, \mathcal{P}) = \sum_{i=1}^n M_i(a_i - a_{i-1})$$

where $m_i = \inf\{f(x) : x \in [a_{i-1}, a_i]\}$, $M_i = \sup\{f(x) : x \in [a_{i-1}, a_i]\}$

Remark 1.3. $m \leq f \leq M, f : [a, b] \rightarrow \mathbb{R}$. For any partition $\mathcal{P}$:

$$U(f, \mathcal{P}) = \sum M_i(a_i - a_{i-1}) \geq m \sum (a_i - a_{i-1}) = m(b - a)$$

$$L(f, \mathcal{P}) = \sum m_i(a_i - a_{i-1}) \leq M \sum (a_i - a_{i-1}) = M(b - a)$$

Hence $\inf_{\mathcal{P}} U(f, \mathcal{P})$ and $\sup_{\mathcal{P}} L(f, \mathcal{P})$ are defined.

Definition 1.4. $f : [a, b] \rightarrow \mathbb{R}$ bounded.

1. Riemann Lower Integral:

$$\underline{\int_a^b} f = \sup_{\mathcal{P}} L(f, \mathcal{P})$$

2. Riemann Upper Integral:

$$\overline{\int_a^b} f = \inf_{\mathcal{P}} U(f, \mathcal{P})$$

3. We say f is Riemann Integrable iff

$$\underline{\int_a^b} f = \overline{\int_a^b} f$$

and we write $\int_a^b f = \underline{\int_a^b} f = \overline{\int_a^b} f$

Proposition 1.5. If $f : [a, b] \rightarrow \mathbb{R}$ is cts, then f is Riemann integrable.

Example 1.6. $f : [0, 1] \rightarrow \mathbb{R}$ via

$$f(x) = \begin{cases} 1, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q} \end{cases}$$

Then $L(f, \mathcal{P}) = 0, U(f, \mathcal{P}) = 1$. Hence $\underline{\int_a^b} f = 0$ and $\overline{\int_a^b} f = 1$. f is not Riemann integrable.

We denote $f := \chi_{\mathbb{Q}}$

Shortfalls of Riemann Integral:

1. Lack of Comparison:

$f = \chi_{\mathbb{Q}} : [0, 1] \rightarrow \mathbb{R}$ is not not Riemann integrable

$|f| \leq 1, 1$ is Riemann integrable

2. Lack of Convergence:

Let $\{q_1, q_2, \dots\}$ be an enumeration of the rationals in $[0, 1]$. Let $f_n = \chi_{\{q_1, \dots, q_n\}}$. Then

$f_n \rightarrow \chi_{\mathbb{Q}}$ pointwisely, (f_n) is increasing, but $\lim_{n \rightarrow \infty} \underbrace{\int_0^1 f_n}_{0} \neq \underbrace{\int_0^1 f}_{\text{DNE}}$.

2 Measures

Goal: Develop a more theoretically rich theory of integration for $f : X \rightarrow \mathbb{R}$. In the case of $X = [a, b]$ and f bounded, we want our new theory to extend Riemann integral.

To Start: To define $\int_X f$, we first make sense of

$$“\mu(X) = \int_X 1”$$

Intuitively, we think of $\mu(X)$, we think of $\mu(X)$ as the size/measure of X .

2.1 Algebras & σ -Algebra

Definition 2.1. X set. An algebra on X is a set $\mathcal{A} \subseteq \mathcal{P}(X)$ such that:

1. $X \in \mathcal{A}$
2. $A \in \mathcal{A} \implies X \setminus A \in \mathcal{A}$
3. $A, B \in \mathcal{A} \implies A \cup B \in \mathcal{A}$

Remark 2.2. $\mathcal{A} \subseteq \mathcal{P}(X)$ algebra, then

1. $\emptyset = X \setminus X \in \mathcal{A}$
2. $A, B \in \mathcal{A}$, then

$$\begin{aligned} X \setminus (A \cap B) &= (X \setminus A) \cup (X \setminus B) \in \mathcal{A} \\ &\implies A \cap B \in \mathcal{A} \end{aligned}$$

3. $A, B \in \mathcal{A}$, $A \setminus B = A \cap (X \setminus B) \in \mathcal{A}$
4. $\mathcal{A}$ closed under finite unions/intersections

Example 2.3. $\mathcal{P}(X) \setminus \{\emptyset, X\}$ algebras.

Example 2.4. $\mathcal{A} = \{A \subseteq X : A \text{ finite, or } X \setminus A \text{ finite}\}$

Definition 2.5. X set. We say $\mathcal{A} \subseteq \mathcal{P}(X)$ is a σ -algebra on X iff

1. $X \in \mathcal{A}$
2. $A \in \mathcal{A} \implies X \setminus A \in \mathcal{A}$
3. $A_1, A_2, A_3, \dots \in \mathcal{A} \implies \bigcup_{n=1}^{\infty} A_n \in \mathcal{A}$

Remark 2.6. $\mathcal{A} \subseteq \mathcal{P}(X)$ is σ -algebra. Then

1. $\emptyset \in \mathcal{A}$
2. $A_i \in \mathcal{A}, \bigcap_{i=1}^n A_i \in \mathcal{A}$

3. $\mathcal{A}$ is an algebra

Idea: We will round up the subsets of X “worth measuring” in a σ -algebra.

Example 2.7. $\mathcal{P}(X), \{\emptyset, X\}$ σ -algebras.

Example 2.8. $\mathcal{A} = \{A \subseteq X : A \text{ countable, or } X \setminus A \text{ countable}\}$ is a σ -algebra.

Definition 2.9. $\mathcal{F} \subseteq \mathcal{P}(X)$. Then

$$\sigma(\mathcal{F}) = \bigcap_{\substack{\mathcal{F} \subseteq \mathcal{A} \\ \mathcal{A} \text{ } \sigma\text{-algebra}}} \mathcal{A}$$

is the σ -algebra generated by $\mathcal{F}$.

Remark 2.10. $\sigma(\mathcal{F})$ is the smallest σ -algebra on X which contains $\mathcal{F}$.

Example 2.11. X topological space (e.g., metric space). $\mathcal{T} = \{A \subseteq X : A \text{ open}\}$. Then, $\sigma(\mathcal{T})$ is called the Borel σ -algebra on X .

Example 2.12. $X = \mathbb{R}$ with standard topology. The following sets are all Borel:

1. open sets
2. closed sets
3. countable sets
4. $(0, 2) \cap [1, 3] = [1, 2)$

Comments: It's hard to imagine the existence of a non-Borel $A \subseteq \mathbb{R}$. The existence of a non-Borel $A \subseteq \mathbb{R}$ requires the axiom of choice.

2.2 Measures

Definition 2.13. $\mathcal{A} \subseteq \mathcal{P}(X)$ a σ -algebra. A measure on $\mathcal{A}$ (OR on X) is a function $\mu : \mathcal{A} \rightarrow [0, \infty]$ such that

1. $\mu(\emptyset) = 0$
2. If $A_1, A_2, A_3, \dots \in \mathcal{A}$ are pairwise disjoint, then

$$\mu\left(\bigsqcup A_i\right) = \sum_{i=1}^{\infty} \mu(A_i)$$

which is well-defined since measure is allowed to be ∞ .

Definition 2.14. X set, $\mathcal{A}$ σ -algebra, μ measure.

$(X, \mathcal{A})$ measurable space
 $(X, \mathcal{A}, \mu)$ measure space

The elements of $\mathcal{A}$ are called measurable sets.

Proposition 2.15 (Monotonicity). $(X, \mathcal{A}, \mu)$. If $A, B \in \mathcal{A}$ and $A \subseteq B$, then

$$\mu(A) \leq \mu(B)$$

Proof. Since $B = A \sqcup (B \setminus A)$, $\mu(B) = \mu(A) + \mu(B \setminus A) \geq \mu(A)$ □

Remark 2.16. Axiom 2 of the definition of μ is called countable additivity. Measures are also finitely additive.

Proposition 2.17 (Excision). $(X, \mathcal{A}, \mu)$. If $A, B \in \mathcal{A}$ with $A \subseteq B$ and $\mu(A) < \infty$, then $\mu(B \setminus A) = \mu(B) - \mu(A)$.

Proof. We have $B = (B \setminus A) \sqcup A \implies \mu(B) = \mu(B \setminus A) + \mu(A)$. □

Proposition 2.18 (Countable Subadditivity). $(X, \mathcal{A}, \mu)$. If $A_1, A_2, A_3, \dots \in \mathcal{A}$, then

$$\mu\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

Proof. Let $B_1 = A_1, B_2 = A_2 \setminus A_1, B_3 = A_3 \setminus (A_1 \cup A_2), \dots$ defined recursively. Then $B_i \in \mathcal{A}$ and $B_i \cap B_j = \emptyset$ for $i \neq j$. Moreover, $\bigcup A_i = \bigsqcup B_i$.

$$\mu\left(\bigcup A_i\right) = \mu\left(\bigsqcup B_i\right) = \sum_{i=1}^{\infty} \mu(B_i) \leq \sum_{i=1}^{\infty} \mu(A_i)$$

□

Proposition 2.19 (Continuity of Measure). $(X, \mathcal{A}, \mu)$. Then

1. If $A_i \in \mathcal{A}$ and $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$, then

$$\mu\left(\bigcup A_i\right) = \lim_{n \rightarrow \infty} \mu(A_n)$$

2. If $B_i \in \mathcal{A}$ and $B_1 \supseteq B_2 \supseteq B_3 \supseteq \dots$, where $\mu(B) < \infty$, then

$$\mu\left(\bigcap B_i\right) = \lim_{n \rightarrow \infty} \mu(B_n)$$

Proof. 1. Let $B_1 = A_1, B_2 = A_2 \setminus A_1, B_3 = A_3 \setminus A_2, \dots$ defined recursively. Therefore

$$\begin{aligned} \mu\left(\bigcup A_i\right) &= \mu\left(\bigsqcup B_i\right) = \sum_{i=1}^{\infty} \mu(B_i) \\ &= \lim_{n \rightarrow \infty} \sum_{i=1}^n \mu(B_i) \\ &= \lim_{n \rightarrow \infty} \mu\left(\bigsqcup_{i=1}^n B_i\right) \\ &= \lim_{n \rightarrow \infty} \mu(A_n) \end{aligned}$$

2. Since $B_1 \supseteq B_2 \supseteq \dots$, then $C_1 \subseteq C_2 \subseteq C_3 \subseteq \dots$, where $C_i = B_1 \setminus B_i$. Hence

$$\begin{aligned}
\mu\left(\bigcap B_i\right) &= \mu(B_1 \setminus \bigcup C_i) \\
&\stackrel{\text{excision}}{=} \mu(B_1) - \mu\left(\bigcup C_i\right) \\
&= \mu(B_1) - \lim_{n \rightarrow \infty} \mu(C_n) \\
&= \mu(B_1) - \lim_{n \rightarrow \infty} \mu(B_1 \setminus B_n) \\
&= \mu(B_1) - \lim_{n \rightarrow \infty} [\mu(B_1) - \mu(B_n)] \\
&= \lim_{n \rightarrow \infty} \mu(B_n)
\end{aligned}$$

□

Example 2.20. $X = \mathbb{N}, \mathcal{A} = \mathcal{P}(\mathbb{N}), \mu(A) = |A|$. $B_n = \{n, n+1, \dots\}$. Then

$$\lim_{n \rightarrow \infty} \mu(B_n) = \infty, \mu\left(\underbrace{\bigcap B_n}_{\emptyset}\right) = 0$$

Example 2.21. $X, \mathcal{A}$ = any σ -algebra. $\mu(A) = |A|$ is counting measure.

Example 2.22. $X, x \in X, \mathcal{A}$ = any σ -algebra

$$\delta_x(A) = \begin{cases} 1 & , x \in A \\ 0 & , x \notin A \end{cases}$$

is the Point-mass Measure w.r.t. x .

Example 2.23. $\mu(A) = \begin{cases} 0 & , A = \emptyset \\ \infty & , A \neq \emptyset \end{cases}$

2.3 Outer Measure

Idea: Given X , start with a reasonable $\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$ which can be used to construct a measure.

Definition 2.24. X set. An outer measure on X is a function

$$\mu^* : \mathcal{P}(X) \rightarrow [0, \infty]$$

such that

1. $\mu^*(\emptyset) = 0$
2. $A \subseteq B \implies \mu^*(A) \leq \mu^*(B)$
3. (Countable Subadditivity)

$$\mu^*\left(\bigcup_{i=1}^{\infty} A_i\right) \leq \sum_{i=1}^{\infty} \mu^*(A_i)$$

Example 2.25. $X = [0, 1], \mu^*(A) = \begin{cases} 1 & , A \text{ uncountable} \\ 0 & , A \text{ countable} \end{cases}$ is an outer measure.

However, $\mu^*([0, 1]) \neq \mu^*([0, \frac{1}{2}]) + \mu^*([\frac{1}{2}, 1])$, so μ^* is NOT a measure.

Definition 2.26. We call $\lambda^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$

$$\lambda^*(A) = \inf \left\{ \sum l(a_i, b_i) : A \subseteq \bigcup (a_i, b_i) \right\}$$

the Lebesgue outer measure on $\mathbb{R}$. Here, $l(a_i, b_i) = b_i - a_i$ is the length of the bounded open interval (a_i, b_i) .

Idea: We know how to measure (a, b) (length). To measure $A \subseteq \mathbb{R}$, we cover A by countably many bounded open intervals and we do it as “finely as possible”.

Proposition 2.27. λ^* is an outer measure.

Proof.

1. Let $\varepsilon > 0$ be arbitrary. Since $\emptyset \subseteq \bigcup (0, \frac{\varepsilon}{2^i})$, then $\lambda^*(\emptyset) \leq \sum_{i=1}^{\infty} \frac{\varepsilon}{2^i} = \varepsilon$, as $\lambda^*(\emptyset)$ is the inf. Hence $\lambda^*(\emptyset) = 0$.
2. Assume $A \subseteq B$. Say $B \subseteq \bigcup I_i$, I_i bounded, open interval. Then $A \subseteq \bigcup I_i$. Hence

$$\begin{aligned} \underbrace{\left\{ \sum l(I_i) : B \subseteq \bigcup I_i \right\}}_X &\subseteq \underbrace{\left\{ \sum l(I_i) : A \subseteq \bigcup I_i \right\}}_Y \\ &\implies \inf Y \leq \inf X \\ &\implies \lambda^*(A) \leq \lambda^*(B) \end{aligned}$$

3. Let $A_i \subseteq \mathbb{R}$ for $i \in \mathbb{N}$. Let $\varepsilon > 0$ be given.
If $\sum \lambda^*(A_i) = \infty$, we are done. Assume $\sum \lambda^*(A_i) < \infty \implies \forall i, \lambda^*(A_i) < \infty$.
There must exist $A_i \subseteq \bigcup_{j=1}^{\infty} (a_{ij}, b_{ij})$ such that

$$\sum_{j=1}^{\infty} l(a_{ij}, b_{ij}) < \lambda^*(A_i) + \frac{\varepsilon}{2^i}$$

Then

$$\begin{aligned} \bigcup_{i=1}^{\infty} A_i &\subseteq \bigcup_{i,j} (a_{ij}, b_{ij}) \\ \implies \lambda^*\left(\bigcup_{i=1}^{\infty} A_i\right) &\leq \sum_i \sum_j l(a_{ij}, b_{ij}) \leq \sum_i \left(\lambda^*(A_i) + \frac{\varepsilon}{2^i}\right) = \sum_i \lambda^*(A_i) + \varepsilon \end{aligned}$$

□

Proposition 2.28. $\lambda^*([a, b]) = b - a$

Proof. Let $\varepsilon > 0$ be given. Since $[a, b] \subseteq (a - \frac{\varepsilon}{2}, b + \frac{\varepsilon}{2})$. Then

$$\lambda^*([a, b]) \leq l(a - \frac{\varepsilon}{2}, b + \frac{\varepsilon}{2}) = b - a + \varepsilon$$

Hence $\lambda^*([a, b]) \leq b - a$. Now suppose $[a, b] \subseteq \bigcup_{i=1}^{\infty} I_i$, I_i bounded open intervals. By compactness,

$$[a, b] \subseteq \sum_{i=1}^N I_i$$

Hence

$$l([a, b]) \leq \sum_{i=1}^N l(I_i) \implies b - a \leq \sum_{i=1}^N l(I_i) \stackrel{\inf}{\implies} b - a \leq \lambda^*([a, b])$$

□

Remark 2.29.

1. $(a, b) \subseteq [a, b] \implies \lambda^*(a, b) \leq b - a$. On the other hand, $[a + \frac{\varepsilon}{2}, b - \frac{\varepsilon}{2}] \subseteq (a, b) \implies b - a - \varepsilon \leq \lambda^*(a, b)$. Therefore $\lambda^*(a, b) = b - a = l(a, b)$.
2. $\forall n, (a, a + n) \subseteq (a, \infty) \implies \forall n, \lambda^*(a, a + n) \leq \lambda^*(a, \infty) \implies \forall n, n \leq \lambda^*(a, \infty) \implies \lambda^*(a, \infty) = \infty$.
3. For all intervals I , $\lambda^*(I) = l(I)$

Definition 2.30. An open rectangle in $\mathbb{R}^n$ is any set of the form

$$R = (a_1, b_1) \times \cdots \times (a_n, b_n)$$

Its volume is $\text{Vol}(R) = l(a_1, b_1) \cdots l(a_n, b_n)$

Proposition 2.31. λ^* is an outer measure and $\lambda^*(R) = \text{Vol}(R)$ for any rectangle.

Idea: Turns out, λ^* is not a measure. To fix this, we restrict the domain of λ^* from $\mathcal{P}(\mathbb{R}^n)$ to a σ -algebra of “reasonable sets”.

Definition 2.32. μ^* outer measure on X . We say B is μ^* -measurable iff for all $A \subseteq X$,

$$\mu^*(A) = \mu^*(A \cap B) + \mu^*(A \cap B^C)$$

Definition 2.33. We say $A \subseteq \mathbb{R}^n$ is Lebesgue measurable iff A is λ^* -measurable.

Remark 2.34. For all $A, B \subseteq X$, we have

$$\begin{aligned} A &= (A \cap B) \cup (A \cap B^C) \\ \implies \mu^*(A) &\leq \mu^*(A \cap B) + \mu^*(A \cap B^C) \end{aligned}$$

So the only interesting part is the $\geq$ part.

Proposition 2.35. Given X, μ^* . If $\mu^*(B) = 0$, then B is μ^* -measurable.

Proof. Suppose $\mu^*(B) = 0$, and let $A \subseteq X$. Then, $A \cap B \subseteq B \implies 0 \leq \mu^*(A \cap B) \leq \mu^*(B) = 0$

$$\implies \mu^*(A \cap B) + \mu^*(A \cap B^C) = \mu^*(A \cap B^C) \leq \mu^*(A)$$

□

Proposition 2.36. $(X = \mathbb{R}, \lambda^*)$. If B is countable, then B is λ^* -measurable (Lebesgue measurable). In fact, $\lambda^*(B) = 0$.

Proof. Say $B = \{b_1, b_2, b_3, \dots\}$, and let $\varepsilon > 0$ be given. Then, each $b_i \in (b_i - \frac{\varepsilon}{2^{i+1}}, b_i + \frac{\varepsilon}{2^{i+1}})$, so

$$\lambda^*(B) \leq \sum \frac{\varepsilon}{2^i} = \varepsilon$$

□

Remark 2.37. Same result follows for $X = \mathbb{R}^n$.

Notation: (X, μ^*) . We denote

$$\mathcal{M}_{\mu^*} = \{B \subseteq X, B \text{ } \mu^* \text{-measurable}\}$$

Theorem 2.38. (X, μ^*) . Then, $\mathcal{M}_{\mu^*}$ is a σ -algebra, and $\mu = \mu^*|_{\mathcal{M}_{\mu^*}}$ is a measure.

Proof. Since $\mu^*(\emptyset) = 0, \emptyset \in \mathcal{M}_{\mu^*}$. Suppose $B \in \mathcal{M}_{\mu^*}$. Then for all $A \subseteq X$,

$$\begin{aligned} \mu^*(A) &= \mu^*(A \cap B) + \mu^*(A \cap B^C) \\ &= \mu^*(A \cap B^C) + \mu^*(A \cap (B^C)^C) \\ &\implies B^C \in \mathcal{M}_{\mu^*} \end{aligned}$$

Claim: $B_1, B_2 \in \mathcal{M}_{\mu^*} \implies B_1 \cup B_2 \in \mathcal{M}_{\mu^*}$.

For all $A \subseteq X$,

$$\begin{aligned} &\mu^*(A \cap (B_1 \cup B_2)) \\ &= \mu^*(A \cap (B_1 \cup B_2) \cap B_1) + \mu^*(A \cap (B_1 \cup B_2) \cap B_1^C) \quad , \text{ by } B_1 \\ &= \mu^*(A \cap B_1) + \mu^*(A \cap B_1^C \cap B_2) \end{aligned}$$

Hence

$$\begin{aligned} &\mu^*(A \cap (B_1 \cup B_2)) + \mu^*(A \cap (B_1 \cup B_2)^C) \\ &= \mu^*(A \cap B_1) + \mu^*(A \cap B_1^C \cap B_2) + \mu^*(A \cap B_1^C \cap B_2^C) \\ &= \mu^*(A \cap B_1) + \mu^*(A \cap B_1^C) \quad , \text{ by } B_2 \\ &= \mu^*(A) \quad , \text{ by } B_1 \end{aligned}$$

Here we finish the first claim.

Claim: If $B_1, B_2, \dots \in \mathcal{M}_{\mu^*}$ are pairwise disjoint, then $\forall A \subseteq X$, we have

$$\mu^*(A) = \left[\sum_{i=1}^n \mu^*(A \cap B_i) \right] + \mu^*(A \cap \left(\bigcap_{i=1}^n B_i^C \right)) \quad (\star)$$

If $n = 1$, we are done. Assume $(\star)$ holds for n . Then

$$\begin{aligned} \mu^*(A \cap \left(\bigcap_{i=1}^n B_i^C \right)) &= \mu^*(A \cap \left(\bigcap_{i=1}^n B_i^C \right) \cap B_{n+1}) + \mu^*(A \cap \left(\bigcap_{i=1}^n B_i^C \right) \cap B_{n+1}^C) \\ &= \mu^*(A \cap B_{n+1}) + \mu^*(A \cap \bigcap_{i=1}^{n+1} B_i^C) \end{aligned}$$

Combining this with $(\star)$,

$$\begin{aligned} \mu^*(A) &= \left[\sum_{i=1}^n \mu^*(A \cap B_i) \right] + \mu^*(A \cap \left(\bigcap_{i=1}^n B_i^C \right)) \\ &= \left[\sum_{i=1}^n \mu^*(A \cap B_i) \right] + \mu^*(A \cap B_{n+1}) + \mu^*(A \cap \bigcap_{i=1}^{n+1} B_i^C) \\ &= \sum_{i=1}^{n+1} \mu^*(A \cap B_i) + \mu^*(A \cap \bigcap_{i=1}^{n+1} B_i^C) \end{aligned}$$

Here we finish the second claim.

By taking $n \rightarrow \infty$ and using monotonicity,

$$\begin{aligned} \mu^*(A) &\geq \left[\sum_{i=1}^{\infty} \mu^*(A \cap B_i) \right] + \mu^*(A \cap \left(\bigcup_{i=1}^{\infty} B_i \right)^C) \quad (\star\star) \\ &\geq \mu^*(A \cap \bigcup_{i=1}^{\infty} B_i) + \mu^*(A \cap \left(\bigcup_{i=1}^{\infty} B_i \right)^C) \\ &\geq \mu^*(A) \end{aligned}$$

Hence $\bigcup B_i \in \mathcal{M}_{\mu^*}$. Thus, if $B_1, B_2, B_3, \dots \in \mathcal{M}_{\mu^*}$,

$$\bigcup B_i = B_1 \sqcup (B_2 \setminus B_1) \sqcup (B_3 \setminus (B_1 \cup B_2)) \sqcup \dots \in \mathcal{M}_{\mu^*}$$

Again, assume $B_1, B_2, \dots$ are pairwise disjoint.

Taking $A = \bigcup B_i$ in $(\star\star)$,

$$\begin{aligned} \mu^*(\bigcup B_i) &= \sum \mu^*(B_i) + \mu^*(\emptyset) \\ &= \sum \mu^*(B_i) \end{aligned}$$

□

Definition 2.39. We call $\lambda : \mathcal{M}_{\lambda^*} \rightarrow [0, \infty]$, $\lambda(B) = \lambda^*(B)$, the Lebesgue measure.

Proposition 2.40. $B = (-\infty, b] \in \mathcal{M}_{\lambda^*}$

Proof. Let $A \subseteq \mathbb{R}$. We must prove that

$$\lambda^*(A) \geq \lambda^*(A \cap B) + \lambda^*(A \cap B^C)$$

We may assume that $\lambda^*(A)$ is finite. Let $\varepsilon > 0$ be given so that there exists

$$A \subseteq \bigcup (a_i, b_i)$$

such that

$$\sum l(a_i, b_i) < \lambda^*(A) + \varepsilon$$

Hence

$$\begin{aligned} & \lambda^*(A \cap B) + \lambda^*(A \cap B^C) \\ & \leq \lambda^*\left(\bigcup (a_i, b_i) \cap B\right) + \lambda^*\left(\bigcup (a_i, b_i) \cap B^C\right) \\ & \leq \sum \lambda^*((a_i, b_i) \cap B) + \sum \lambda^*((a_i, b_i) \cap B^C) \\ & = \sum [l((a_i, b_i) \cap B) + l((a_i, b_i) \cap B^C)] \\ & = \sum l(a_i, b_i) \\ & < \lambda^*(A) + \varepsilon \end{aligned}$$

□

Investigation: $a \leq b$, then

$$(-\infty, b] \cap (-\infty, a]^C = (-\infty, b] \cap (a, \infty) = (a, b] \in \mathcal{M}_{\lambda^*}$$

$$\implies (a, b) = \bigcup_{n=1}^{\infty} (a, b - \frac{1}{n}] \in \mathcal{M}_{\lambda^*}$$

Therefore $U \subseteq \mathbb{R} \implies U$ is Lebesgue measurable. Thus, $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{M}_{\lambda^*}$.

Proposition 2.41. $\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}_{\lambda^*}$

Question: Does there exist $A \in \mathcal{M}_{\lambda^*}$ such that $A \notin \mathcal{B}(\mathbb{R})$? **Answer:** Yes! Prove later.

Remark 2.42. $\lambda : \mathcal{B}(\mathbb{R}^n) \rightarrow [0, \infty]$ is also called Lebesgue measure. In this class, we assume the domain of λ is $\mathcal{M}_{\lambda^*}$.

Investigation: Assume $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu)$ is a measure space such that $\mu(A) < \infty$ for all compact $K \subseteq \mathbb{R}$. Consider $F : \mathbb{R} \rightarrow \mathbb{R}$ via

$$F(x) = \begin{cases} \mu([0, x]) & x \geq 0 \\ -\mu((x, 0)) & x < 0 \end{cases}$$

Note that F is increasing. Assume $x_n \searrow x$ (i.e., $x_n \rightarrow x$, (x_n) decreases).

1. $x \geq 0$, then

$$F(x) = \mu([0, x]) = \mu\left(\bigcap [0, x_n]\right) \stackrel{\text{CM}}{=} \lim_{n \rightarrow \infty} \mu([0, x_n]) = \lim_{n \rightarrow \infty} F(x_n)$$

2. Similarly, if $x < 0$ and $x_n \searrow x$, then

$$\lim_{n \rightarrow \infty} F(x_n) = F(x)$$

Hence F is right continuous.

Let $F : \mathbb{R} \rightarrow \mathbb{R}$ be increasing and right continuous. For $a \leq b$, let

$$l_F(a, b] = F(b) - F(a)$$

Then:

1. $\mu_F^* : \mathcal{P}(\mathbb{R}) \rightarrow [0, \infty]$ with

$$\mu_F^*(A) = \inf\left\{\sum l_F(a_i, b_i] : A \subseteq \bigcup (a_i, b_i]\right\}$$

is an outer measure.

2. $\mu_F = \mu_F^*|_{\mathcal{M}_{\mu_F^*}}$ is a measure. Lebesgue-Stieltjes measure

Example 2.43. $\mu_F = \lambda, F(x) = x$

2.4 Lebesgue Measure

Proposition 2.44. $A \subseteq \mathbb{R}^n$ be Lebesgue measurable. Then:

1. $\lambda(A) = \inf\{\lambda(U) : A \subseteq U \text{ open}\}$
2. $\lambda(A) = \sup\{\lambda(K) : K \subseteq A \text{ compact}\}$

Proof. 1. If $A \subseteq U$ open, $\lambda(A) \leq \lambda(U) \implies \lambda(A) \leq \inf\{\lambda(U)\}$. We may assume that $\lambda(A) < \infty$. Let $\varepsilon > 0$ be given so that we may find open rectangles R_i with

$$\sum \text{Vol}(R_i) < \lambda(A) + \varepsilon, \text{ and } A \subseteq \bigcup R_i$$

Consider $U = \bigcup R_i$. Hence

$$\begin{aligned} \lambda(U) &\leq \sum \lambda(R_i) = \sum \text{Vol}(R_i) < \lambda(A) + \varepsilon \\ &\implies \inf\{\lambda(U)\} \leq \lambda(U) < \lambda(A) + \varepsilon \end{aligned}$$

2. We split it into two cases:

(i) A is bounded.

Say $A \subseteq [-N, N]^n = C$. Let $\varepsilon > 0$ be given. By 1, we may find an open set $C \setminus A \subseteq U$ such that $\lambda(U) < \lambda(C \setminus A) + \varepsilon$. Hence

$$\lambda(U) < \lambda(C) - \lambda(A) + \varepsilon \implies \lambda(A) < \lambda(C) - \lambda(U) + \varepsilon$$

Let $K = C \setminus U$, which is compact. Next, $K \subseteq A$. Moreover,

$$C \subseteq K \sqcup U$$

$$\implies \lambda(C) \leq \lambda(K) + \lambda(U)$$

$$\implies \lambda(A) < \lambda(C) - \lambda(U) + \varepsilon \leq \lambda(K) + \varepsilon$$

$$\implies \lambda(A) \leq \sup\{\lambda(K) : K \subseteq A \text{ compact}\}$$

Clearly,

$$\lambda(A) \geq \sup\{\lambda(K) : K \subseteq A \text{ compact}\}$$

(ii) A is not bounded.

Let $b \in \mathbb{R}$ such that $b < \lambda(A)$. We produce a compact $K \subseteq A$ such that $b < \lambda(K)$. Let $A_n = B_n(0) \cap A$ so that A_n is bounded and measurable, $A = \bigcup A_n$, $A_1 \subseteq A_2 \subseteq A_3 \subseteq \dots$

$$\implies \lambda(A) \stackrel{\text{CM}}{=} \lim \lambda(A_n)$$

Hence there exists n such that $\lambda(A_n) > b$. By Case (i), there exists compact $K \subseteq A_n$ such that $\lambda(K) > b$.

□

Lemma 2.45. $U \subseteq \mathbb{R}^n$ open. Then, U is the countable, disjoint union of half-open cubes. (i.e., $[a_1, b_1) \times [a_2, b_2) \times \dots \times [a_n, b_n)$)

Why? Let $\mathcal{C}_k :=$ cubes of the form $\{x \in \mathbb{R}^n : \frac{J_i}{2^k} \leq x_i < \frac{J_i+1}{2^k}\}$, where $J_1, \dots, J_n \in \mathbb{Z}$. Then

1. The elements of $\mathcal{C}_k$ partition $\mathbb{R}^n$, $\mathcal{C}_k$ is countable.

2. If $k_1 < k_2$, then each cube in $\mathcal{C}_{k_2}$ is contained in some cubes in $\mathcal{C}_{k_1}$

Let $U \subseteq \mathbb{R}^n$ be open.

$$\mathcal{D}_0 = \emptyset$$

$$\mathcal{D}_1 = \text{cubes in } \mathcal{C}_1, \subseteq U$$

$$\mathcal{D}_2 = \text{cubes in } \mathcal{C}_2, \subseteq U, \text{ disjoint from previous cubes}$$

$$\vdots$$

$$\mathcal{D}_k = \text{cubes in } \mathcal{C}_k, \subseteq U, \text{ disjoint from previous cubes}$$

$$\mathcal{D} = \bigcup_{k=1}^{\infty} \mathcal{D}_k$$

Clearly, $\bigcup_{D \in \mathcal{D}} D \subseteq U$. Let $x \in U$. Since U is open, there exists a minimal k_0 such that $x \in$ a cube in $\mathcal{D}_{k_0}$, hence $x \in \mathcal{D}$.

Proposition 2.46. Lebesgue measure is the only measure on $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ which assigns each rectangle its volume.

Proof. We already know that λ is such a measure. Let μ be another such measure. Let $U \subseteq \mathbb{R}^n$ be open, so that

$$U = \bigsqcup C_i$$

where C_i is a cube. Then

$$\lambda(U) = \sum \lambda(C_i) = \sum \mu(C_i) = \mu(U)$$

Let $A \in \mathcal{B}(\mathbb{R}^n)$. If $A \subseteq U$ is open, then

$$\mu(A) \leq \mu(U) = \lambda(U)$$

$$\implies \mu(A) \leq \inf\{\lambda(U) : A \subseteq U \text{ open}\} = \lambda(A)$$

(i) A is bounded.

Let V be open and bounded such that $A \subseteq V$. Then

$$\begin{aligned} \mu(V) &= \mu(A) + \mu(V \setminus A) \\ &\leq \lambda(A) + \lambda(V \setminus A) \\ &= \lambda(V) = \mu(V) \end{aligned}$$

Since $\mu(A) \leq \lambda(A)$ and $\mu(V \setminus A) \leq \lambda(V \setminus A)$, we conclude that

$$\lambda(A) = \mu(A)$$

(ii) A is not bounded.

Let $B_n = \{x \in \mathbb{R}^n : n-1 \leq \|x\| < n\}$, $A = \bigsqcup (A \cap B_n)$. Hence,

$$\lambda(A) = \sum \lambda(A \cap B_n) = \sum \mu(A \cap B_n) = \mu(A)$$

□

Definition 2.47. $A \subseteq \mathbb{R}^n, x \in \mathbb{R}^n$. We call

$$x + A := \{x + a : a \in A\}$$

the translation of A by x .

Proposition 2.48. $A \subseteq \mathbb{R}^n, x \in \mathbb{R}^n$. Then

1. $\lambda^*(A) = \lambda^*(x + A)$
2. $B \in \mathcal{M}_{\lambda^*}$, then $x + B \in \mathcal{M}_{\lambda^*}$

Proof.

1. Note that

$$A \subseteq \bigcup R_i \iff x + A \subseteq \bigcup (x + R_i)$$

$$\implies \{\sum \text{Vol}(R_i) : A \subseteq \bigcup R_i\} = \{\sum \text{Vol}(R_i) : x + A \subseteq \bigcup R_i\}$$

2. Let $B \in \mathcal{M}_{\lambda^*}$ and let $A \subseteq \mathbb{R}^n$. Then,

$$\begin{aligned} & \lambda^*((x + B) \cap A) + \lambda^*((x + B)^C \cap A) \\ &= \lambda^*(B \cap (-x + A) + x) + \lambda^*(B^C \cap (-x + A) + x) \\ &= \lambda^*(B \cap (-x + A)) + \lambda^*(B^C \cap (-x + A)) \\ &= \lambda^*(-x + A) \\ &= \lambda^*(A) \end{aligned}$$

□

Recall:

$$\mathcal{B}(\mathbb{R}^n) \subseteq \mathcal{M}_{\lambda^*} \subseteq \mathcal{P}(\mathbb{R}^n)$$

Goals:

1. Construct a set which is not Lebesgue measurable.
2. Construct a Lebesgue measurable set which is not Borel.

Lemma 2.49. $A \subseteq \mathbb{R}$ bounded and measurable, $X \subseteq \mathbb{R}$ bounded and countably infinite. If $x + A$ ($x \in X$) are pairwise disjoint, then $\lambda(A) = 0$

Proof. We have

$$\lambda(\underbrace{\bigsqcup_x (x + A)}_{< \infty}) = \sum \lambda(x + A) = \sum_x \lambda(A)$$

$$\implies \lambda(A) = 0.$$

□

Construction: $A \subseteq \mathbb{R}$. $a, b \in A$. We say $a \sim b \iff a - b \in \mathbb{Q}$ (equivalence relation). Let $\mathcal{C}_A$ denote a single choice of equivalence class representatives.

Remark 2.50. For $x \neq y \in \mathbb{Q}$, we have

$$(x + \mathcal{C}_A) \cap (y + \mathcal{C}_A) = \emptyset$$

Proof. Suppose for contradiction that $x + z_1 = y + z_2, z_1, z_2 \in \mathcal{C}_A$. Then

$$x - y = z_2 - z_1 \in \mathbb{Q} \implies z_1 \sim z_2 \implies z_1 = z_2 \implies x = y$$

□

Theorem 2.51 (Vitali). Every $A \subseteq \mathbb{R}$ with $\lambda^*(A) > 0$ contains a non-Lebesgue measurable set.

Proof.

1. A is bounded. Say $A \subseteq [-N, N]$.

Claim: $\mathcal{C}_A$ is not Lebesgue measurable.

Claim. For $x \in \mathbb{Q} \cap (0, 1)$, $x + \mathcal{C}_A$ are pairwise disjoint. Assume $\mathcal{C}_A$ is Lebesgue measurable. By the lemma, $\lambda(\mathcal{C}_A) = 0$. Let $a \in A$ so that $a \sim b \in \mathcal{C}_A$

$$\implies a - b = q \in \mathbb{Q} \implies q \in [-2N, 2N] \cap \mathbb{Q}$$

Taking $Q = \mathbb{Q} \cap [-2N, 2N]$, then

$$\begin{aligned} A &\subseteq \bigcup_{q \in Q} q + \mathcal{C}_A \\ \implies \lambda^*(A) &\leq \sum \underbrace{\lambda(q + \mathcal{C}_A)}_0 = 0 \end{aligned}$$

Contradiction. Therefore $\mathcal{C}_A$ is not Lebesgue measurable. □

2. A not bounded. Let $B_n = \{x \in \mathbb{R} : n - 1 \leq |x| < n\} \in \mathcal{B}(\mathbb{R})$. Then

$$A = \bigsqcup (A \cap B_n) \implies \exists n, \lambda^*(A \cap B_n) > 0$$

Apply Case 1 to $A \cap B_n$. □

The non-Lebesgue measurable subset $\mathcal{C}_A \subseteq A$ is called the Vitali subset of A .

Corollary 2.52. There exists $A, B \subseteq \mathbb{R}$ such that $A \cap B = \emptyset$ and

$$\lambda^*(A \cup B) < \lambda^*(A) + \lambda^*(B)$$

Proof. $C \subseteq \mathbb{R}$ non-Lebesgue measurable.

$$\implies \exists X \subseteq \mathbb{R}, \lambda^*(\underbrace{X}_{A \cup B}) < \lambda^*(\underbrace{X \cap C}_A) + \lambda^*(\underbrace{X \cap C^C}_B)$$

□

2.5 Completeness

Definition 2.53. $(X, \mathcal{A}, \mu)$. We say $(X, \mathcal{A}, \mu)$ (or just μ) is complete iff

$$\forall A \in \mathcal{A} \text{ with } \mu(A) = 0, \text{ then } B \subseteq A \implies B \in \mathcal{A}$$

Example 2.54. μ^* outer measure on X , then $(X, \mathcal{M}_{\mu^*}, \mu)$ is complete.

Definition 2.55. $(X, \mathcal{A}, \mu)$. Then completion of $\mathcal{A}$ under μ is

$$\mathcal{A}_\mu := \{A \subseteq X : \exists E, F \in \mathcal{A}, E \subseteq A \subseteq F, \mu(F \setminus E) = 0\}$$

Remark 2.56 $(\star)$.

1. $A \in \mathcal{A}$, take $E = F = A \implies A \in \mathcal{A}_\mu$. Thus, $\mathcal{A} \subseteq \mathcal{A}_\mu$.
2. $(X, \mathcal{A}, \mu)$ complete.
 $A \in \mathcal{A}_\mu$, then $\exists E, F \in \mathcal{A}, E \subseteq A \subseteq F, \mu(F \setminus E) = 0$. Therefore

$$\begin{aligned} A &= \underbrace{(A \setminus E)}_{\subseteq F \setminus E} \cup E \\ &\implies A \in \mathcal{A}_\mu \end{aligned}$$

3. $A \in \mathcal{A}_\mu, E \subseteq A \subseteq F, \mu(F \setminus E) = 0$.

- (a) If $\mu(E) = \infty$, then $\mu(F) = \infty$
- (b) If $\mu(E) < \infty, \mu(F) - \mu(E) = 0 \implies \mu(E) = \mu(F)$. If $B \in \mathcal{A}$ with $B \subseteq A$, then

$$\mu(B) \leq \mu(F) = \mu(E)$$

$$\implies \mu(E) = \sup\{\mu(B) : B \in \mathcal{A}, B \subseteq A\} \text{ (RHS only depends on } \mu \text{ and } A)$$

Proposition 2.57. $(X, \mathcal{A}, \mu)$. Then $(X, \mathcal{A}_\mu, \bar{\mu})$ is a complete measure space, where

$$\bar{\mu} = \sup\{\mu(B) : B \in \mathcal{A}, B \subseteq A\}$$

Also, $\bar{\mu}|_{\mathcal{A}} = \mu$.

Proof.

Claim 1: $\mathcal{A}_\mu$ is a σ -algebra.

Claim 1. Since $X \in \mathcal{A}, X \in \mathcal{A}_\mu$.

Assume $A \in \mathcal{A}_\mu$ so that $\exists E, F \in \mathcal{A}$ with $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$. Then

$$F^C \subseteq A^C \subseteq E^C, \text{ and } \mu(E^C \setminus F^C) = \mu(F \setminus E) = 0$$

Hence $A^C \in \mathcal{A}_\mu$.

Let $A_n \in \mathcal{A}_\mu$ for $n \in \mathbb{N}$ and take $E_n, F_n \in \mathcal{A}, E_n \subseteq A_n \subseteq F_n, \mu(F_n \setminus E_n) = 0$. Then

$$\begin{aligned} \bigcup E_n &\subseteq \bigcup A_n \subseteq \bigcup F_n \\ \implies \mu((\bigcup F_n) \setminus (\bigcup E_n)) &\leq \mu(\bigcup (F_n \setminus E_n)) \leq \sum \mu(F_n \setminus E_n) = 0 \end{aligned}$$

Hence $\bigcup A_n \in \mathcal{A}_\mu$. □

Claim 2: $\bar{\mu}$ is a measure with $\bar{\mu}|_{\mathcal{A}} = \mu$.

Claim 2. If $A \in \mathcal{A}$, then $\bar{\mu}(A) = \mu(A)$ by taking $E = A$ (last Remark). In particular, $\bar{\mu}(\emptyset) = \mu(\emptyset) = 0$.

Let $A_n \in \mathcal{A}_\mu$ be pairwise disjoint. Taken $E_n \subseteq A_n \subseteq F_n$ with $\mu(F_n \setminus E_n) = 0$.

$$\implies \bigsqcup E_n \subseteq \bigsqcup A_n \subseteq \bigcup F_n$$

Again, $\mu((\bigcup F_n) \setminus (\bigcup E_n)) = 0$, and so

$$\bar{\mu}(\bigsqcup A_n) \stackrel{\text{Remark}}{=} \mu(\bigsqcup E_n) = \sum \mu(E_n) = \sum \bar{\mu}(A_n)$$

□

Claim 3: $\bar{\mu}$ is complete.

Claim 3. Let $A \in \mathcal{A}_\mu$ such that $\bar{\mu}(A) = 0$. Take $E \subseteq A \subseteq F$, $\mu(F \setminus E) = 0$. Then

$$\mu(F) = \mu(E) = \bar{\mu}(A) = 0$$

If $B \subseteq A$, then $\emptyset \subseteq B \subseteq F$ with $\mu(F \setminus \emptyset) = \mu(F) = 0 \implies B \in \mathcal{A}_\mu$.

□

□

Lemma 2.58. $A \in \mathcal{M}_{\lambda^*}$. There exists $E \subseteq A \subseteq F$, $E, F \in \mathcal{B}(\mathbb{R}^n)$ such that $\lambda(F \setminus E) = 0$

Proof. For $\lambda(A) < \infty$, we can find $K_n \subseteq A \subseteq U_n$, with K_n compact and U_n open, such that

$$\lambda(U_n) < \lambda(A) + \frac{1}{n}, \lambda(A) - \frac{1}{n} < \lambda(K_n).$$

Let $E = \bigcup K_n$, $F = \bigcap U_n$, $E \subseteq A \subseteq F$. Also,

$$\lambda(F \setminus E) \leq \lambda(U_n \setminus K_n) = \lambda(U_n \setminus A) + \lambda(A \setminus K_n) < \frac{2}{n} \rightarrow 0.$$

Assume $\lambda(A) = \infty$. Take $A_n := A \cap \{x \in \mathbb{R}^n : n-1 \leq \|x\| < n\}$. Then $A = \bigsqcup A_n$, and we may find

$$E_n \subseteq A_n \subseteq F_n \text{ borel, such that } \lambda(F_n \setminus E_n) = 0.$$

Then $\bigcup E_n \subseteq A \subseteq \bigcup F_n$, and

$$\lambda((\bigcup F_n) \setminus (\bigcup E_n)) \leq \lambda(\bigcup (F_n \setminus E_n)) \leq \sum \lambda(F_n \setminus E_n) = 0.$$

□

Proposition 2.59. $(\mathbb{R}^n, \mathcal{M}_{\lambda^*}, \lambda)$ is the completion of $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \lambda)$.

Later: $(\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n), \lambda)$ is not complete.

Proof. Let $\mu :=$ Lebesgue measure on the $\mathcal{B}(\mathbb{R}^n)$. If $A \in \mathcal{M}_{\lambda^*}$, then $A \in \mathcal{B}(\mathbb{R}^n)_\mu$ by the lemma. Let $A \in \mathcal{B}(\mathbb{R}^n)_\mu$. Then there exists $E, F \in \mathcal{B}(\mathbb{R}^n)$ such that $E \subseteq A \subseteq F$, and $\mu(F \setminus E) = 0$. Since λ is complete, $A \setminus E$ is Lebesgue measurable. Then

$$A = (A \setminus E) \cup E \in \mathcal{M}_{\lambda^*}.$$

□

Definition 2.60. $(X, \mathcal{A}, \mu)$. Then the outer measure induced by μ is

$$\mu^*(A) = \inf\{\mu(B) : A \subseteq B, B \in \mathcal{A}\}.$$

Proposition 2.61. $(X, \mathcal{A}, \mu)$. The induced outer measure μ^* is an outer measure.

Proof. Since $\emptyset \subseteq \emptyset$, then $\mu^*(\emptyset) \leq \mu(\emptyset) = 0 \implies \mu^*(\emptyset) = 0$.

Suppose $A \subseteq B \subseteq X$. Then

$$\{\mu(C) : C \in \mathcal{A}, B \subseteq C\} \subseteq \{\mu(C) : C \in \mathcal{A}, A \subseteq C\}$$

$$\implies \mu^*(A) \leq \mu^*(B).$$

Let $A_n \subseteq X$ for every $n \in \mathbb{N}$. We show

$$\mu^*\left(\bigcup A_n\right) \leq \sum \mu^*(A_n)$$

Let $\varepsilon > 0$ be given. We may assume $\mu^*(A_n) < \infty, \forall n \in \mathbb{N}$. For each n , we may find $B_n \in \mathcal{A}, A_n \subseteq B_n$ such that

$$\mu^*(B_n) < \mu^*(A_n) + \frac{\varepsilon}{2^n}$$

Therefore $\bigcup A_n \subseteq \bigcup B_n$, and so

$$\mu^*\left(\bigcup A_n\right) \leq \mu\left(\bigcup B_n\right) \leq \sum \mu(B_n) \leq \sum [\mu^*(A_n) + \frac{\varepsilon}{2^n}] = \sum \mu^*(A_n) + \varepsilon$$

□

HW: $\mu^*|_{\mathcal{A}} = \mu$.

Definition 2.62. $(X, \mathcal{A}, \mu)$. Then inner measure induced by μ is

$$\mu_*(A) = \sup\{\mu(B) : B \in \mathcal{A}, B \subseteq A\}.$$

Proposition 2.63. $\mu_*(A) \leq \mu^*(A)$.

Proof. If $B \in \mathcal{A}$ and $B \subseteq A$, then

$$\mu(B) = \mu^*(B) \leq \mu^*(A)$$

$$\implies \mu_*(A) = \sup\{\mu(B) : B \in \mathcal{A}, B \subseteq A\} \leq \mu^*(A)$$

□

Proposition 2.64. $(X, \mathcal{A}, \mu), \mu^*(A) < \infty$. Then, $A \in \mathcal{A}_\mu$ iff $\mu_*(A) = \mu^*(A)$.

Proof. Let $A \in \mathcal{A}_\mu$ so that there exists $E, F \in \mathcal{A}$ such that $E \subseteq A \subseteq F$ and $\mu(F \setminus E) = 0$. Then,

$$\mu(E) \leq \mu_*(A) \leq \mu^*(A) \leq \mu(F)$$

Since $\mu(E) = \mu(F)$, we are done.

Now let's assume $\mu_*(A) = \mu^*(A)$. For $n \in \mathbb{N}$, there exists $E_n, F_n \in \mathcal{A}$ such that

$$E_n \subseteq A \subseteq F_n$$

and

$$\mu_*(A) - \frac{1}{n} < \mu(E_n), \quad \mu(F_n) < \mu^*(A) + \frac{1}{n}$$

Take $E = \bigcup E_n$ and $F = \bigcap F_n$. Then $E \subseteq A \subseteq F$ and

$$\begin{aligned} \mu(F \setminus E) &\leq \mu(F_n \setminus E_n) \\ &= \mu(F_n) - \mu(E_n) \\ &< \mu^*(A) + \frac{1}{n} - (\mu_*(A) - \frac{1}{n}) \\ &= \frac{2}{n} \longrightarrow 0. \end{aligned}$$

□

Proposition 2.65. $(\mathbb{R}^n, \mathcal{M}_{\lambda^*}, \lambda)$.

$$\lambda^*(A) = \underbrace{\inf\{\lambda(B) : A \subseteq B, B \in \mathcal{M}_{\lambda^*}\}}_{\mu^*(A)}$$

Proof. If $B \in \mathcal{M}_{\lambda^*}$ and $A \subseteq B$, then $\lambda^*(A) \leq \lambda(B)$.

$$\implies \lambda^*(A) \leq \mu^*(A)$$

Let $A \subseteq \bigcup R_i$. Then

$$\mu^*(A) \leq \sum \mu^*(R_i) = \sum \lambda(R_i)$$

(the first $\leq$ is by monotonicity, the second $=$ is by $\mu^*|_{\mathcal{A}} = \mu$ from **HW**)

$$\implies \mu^*(A) \leq \lambda^*(A)$$

□

Corollary 2.66. $A \subseteq \mathbb{R}^n, \lambda(A) < \infty$. Then $A \in \mathcal{M}_{\lambda^*}$ iff $\lambda_*(A) = \lambda^*(A)$.

Example 2.67. $V \subseteq (0, 1)$ Vitali. Let $A = V \cup (2, \infty) \notin \mathcal{M}_{\lambda^*}$. Then $\lambda_*(A) = \lambda^*(A) = \infty$.

3 Functions and Integrals

3.1 Measurable Functions

Proposition 3.1. $(X, \mathcal{A}), A \in \mathcal{A}, f : A \rightarrow [-\infty, \infty]$. TFAE:

- a) $\forall \alpha \in \mathbb{R}, \{x \in A : f(x) \leq \alpha\} = f^{-1}[-\infty, \alpha] \in \mathcal{A}$
- b) $\forall \alpha \in \mathbb{R}, \{x \in A : f(x) < \alpha\} = f^{-1}[-\infty, \alpha) \in \mathcal{A}$
- c) $\forall \alpha \in \mathbb{R}, \{x \in A : f(x) \geq \alpha\} = f^{-1}[\alpha, \infty] \in \mathcal{A}$
- d) $\forall \alpha \in \mathbb{R}, \{x \in A : f(x) > \alpha\} = f^{-1}(\alpha, \infty] \in \mathcal{A}$

Proof. $[-\infty, \alpha) = \bigcup_n [-\infty, \alpha - \frac{1}{n}]$. Then $f^{-1}[-\infty, \alpha) = \bigcup f^{-1}[-\infty, \alpha - \frac{1}{n}]$.
Also, $[-\infty, \alpha] = \bigcap [-\infty, \alpha + \frac{1}{n}]$. Then $f^{-1}[-\infty, \alpha] = \bigcap f^{-1}[-\infty, \alpha + \frac{1}{n}]$
c), d) take complements of b), a). □

Definition 3.2. $(X, \mathcal{A}), A \in \mathcal{A}, f : A \rightarrow [-\infty, \infty]$. We say f is $(\mathcal{A}-)$ measurable iff $f^{-1}[-\infty, \alpha) \in \mathcal{A}, \forall \alpha \in \mathbb{R}$.

Remark 3.3. $X = \mathbb{R}^n$. Then

- 1. $A = \mathcal{M}_{\lambda^*}$, measurable = Lebesgue measurable.
- 2. $\mathcal{A} = \mathcal{B}(\mathbb{R}^n)$, measurable = Borel measurable.

Proposition 3.4. $(X, \mathcal{A}), A \in \mathcal{A}, f : A \rightarrow \mathbb{R}$. Then, f is measurable iff $f^{-1}(U) \in \mathcal{A}$ for all $U \subseteq \mathbb{R}$ open.

Proof. ($\Leftarrow$) Take $\alpha \in \mathbb{R}$. Then

$$f^{-1}[-\infty, \alpha) = \bigcup_{n=1}^{\infty} f^{-1}(\underbrace{(\alpha - n, \alpha)}_{\in \mathcal{A}}) \in \mathcal{A}.$$

($\Rightarrow$) Notice $(a, b) = [-\infty, b) \cap (a, \infty]$. Then

$$f^{-1}(a, b) = f^{-1}[-\infty, b) \cap f^{-1}(a, \infty] \in \mathcal{A}$$

Thus, $\forall U \subseteq \mathbb{R}$ open, $f^{-1}(U) \in \mathcal{A}$. □

Remark 3.5. $f : A \rightarrow \mathbb{R}$ measurable. By A1Q2, $f^{-1}(B) \in \mathcal{A}$ for any Borel set $B \subseteq \mathbb{R}$.

Example 3.6. $A \in \mathcal{M}_{\lambda^*}, f : A \rightarrow \mathbb{R}$ continuous, then f is Lebesgue measurable. In fact, f is Borel measurable.

Example 3.7. Borel measurable $\Rightarrow$ Lebesgue measurable.

Example 3.8. $f : I \rightarrow \mathbb{R}$ (weakly) increasing, I interval. Then $f^{-1}[-\infty, \alpha)$ is an interval, therefore f is Borel measurable.

Example 3.9 ($\star$). $(X, \mathcal{A})$, $A \in \mathcal{A}$. Then the characteristic function of A is $\chi_A : X \rightarrow [-\infty, \infty]$ via

$$\chi_A(x) = \begin{cases} 1 & , x \in A \\ 0 & , x \notin A \end{cases}$$

Then $\chi_A^{-1}[-\infty, \alpha] \in \{X, A, X \setminus A, \emptyset\}$

Proposition 3.10. $(X, \mathcal{A})$, $A \in \mathcal{A}$, $f, g : A \rightarrow [-\infty, \infty]$ measurable. Then

1. $\{x \in A : f(x) < g(x)\} \in \mathcal{A}$
2. $\{x \in A : f(x) \leq g(x)\} \in \mathcal{A}$
3. $\{x \in A : f(x) = g(x)\} \in \mathcal{A}$

Proof. If f or g is constant, then we are done. We may now assume that f and g are both not constant.

1. Notice that

$$\{x \in A : f(x) < g(x)\} = \bigcup_{q \in \mathbb{Q}} (\underbrace{\{x \in A : f(x) < q\}}_{\in \mathcal{A}} \cap \underbrace{\{x \in A : q < g(x)\}}_{\in \mathcal{A}})$$

2. By using (1),

$$\{x \in A : f(x) \leq g(x)\} = A \setminus \{x \in A : g(x) < f(x)\} \in \mathcal{A}$$

3. By using (2),

$$\{x \in A : f(x) = g(x)\} = \{x \in A : f(x) \leq g(x)\} \cap \{x \in A : g(x) \leq f(x)\} \in \mathcal{A}$$

□

Definition 3.11. We write

$$(f \vee g)(x) = \max\{f(x), g(x)\}$$

$$(f \wedge g)(x) = \min\{f(x), g(x)\}$$

Proposition 3.12. $f, g : A \rightarrow [-\infty, \infty]$ measurable, then $f \wedge g, f \vee g$ are measurable.

Proof.

$$\{x \in A : (f \vee g)(x) \leq \alpha\} = \{x \in A : f(x) \leq \alpha\} \cap \{x \in A : g(x) \leq \alpha\}$$

$$\{x \in A : (f \wedge g)(x) \leq \alpha\} = \{x \in A : f(x) \leq \alpha\} \cup \{x \in A : g(x) \leq \alpha\}$$

□

Proposition 3.13. $f : A \rightarrow \mathbb{R}$ measurable, $g : \mathbb{R} \rightarrow \mathbb{R}$ continuous. Then $g \circ f$ is measurable.

Proof. Let $U \subseteq \mathbb{R}$ open, then

$$(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)) \text{ is measurable}$$

□

Corollary 3.14. $f : A \rightarrow \mathbb{R}$ measurable. Then $\alpha f, f^2, |f|$ are measurable.

Proposition 3.15. $f, g : A \rightarrow \mathbb{R}$ measurable. Then $f + g$ is measurable.

Proof. $\{x \in A : f(x) + g(x) < \alpha\} = \{x \in A : f(x) < \underbrace{\alpha - g(x)}_{h(x)}\} \in \mathcal{A}$, where $h(x) = (\alpha - x) \circ g$. □

Proposition 3.16. $f, g : A \rightarrow \mathbb{R}$ measurable. Then fg is measurable.

Proof. $(f + g)^2 = \underbrace{f^2}_{\text{meas}} + 2fg + \underbrace{g^2}_{\text{meas}}$ measurable, so $2fg$ is measurable, and therefore fg is measurable. □

HW: $f : A \rightarrow \mathbb{R}$ measurable, $\forall x, f(x) \neq 0$. Prove that $\frac{1}{f}$ is measurable.

Proposition 3.17. $f_n : A \rightarrow [-\infty, \infty]$ measurable, then

a) $\sup f_n : x \mapsto \sup\{f_n(x) : n \in \mathbb{N}\}$ is measurable.

b) $\inf f_n$ is measurable.

Proof.

$$\begin{aligned} \{x \in A : \sup f_n(x) \leq \alpha\} &= \bigcap_{n=1}^{\infty} \underbrace{\{x \in A : f_n(x) \leq \alpha\}}_{\in \mathcal{A}} \\ \{x \in A : \inf f_n(x) \geq \alpha\} &= \bigcap_{n=1}^{\infty} \underbrace{\{x \in A : f_n(x) \geq \alpha\}}_{\in \mathcal{A}} \end{aligned}$$

□

Proposition 3.18. $f_n : A \rightarrow \mathbb{R}$ measurable. Then

$$\limsup f_n : x \mapsto \limsup\{f_n(x) : n \in \mathbb{N}\}$$

is measurable and $\liminf f_n$ is measurable.

Proof. Notice that

$$\limsup\{f_n(x) : n \in \mathbb{N}\} = \lim_{n \rightarrow \infty} (\sup\{f_n(x) : n \geq k\}) = \inf_k \sup_{n \geq k} f_n(x)$$

Similarly,

$$\liminf f_n = \sup_k \inf_{n \geq k} f_n$$

□

Proposition 3.19. $f_n : A \rightarrow \mathbb{R}$ measurable. $B = \{x \in A : \limsup f_n(x) = \liminf f_n(x)\}$. Then $\lim f_n$ is a measurable function on B .

Proof.

$$\{x \in B : \lim f_n(x) \leq \alpha\} = B \cap \{x \in A : \limsup f_n(x) \leq \alpha\} \in \mathcal{A}.$$

□

Corollary 3.20. If $f_n : A \rightarrow \mathbb{R}$ is measurable and $f_n \rightarrow f$ pointwisely, then f is measurable.

Definition 3.21. $f : A \rightarrow [-\infty, \infty]$, we call

$$f^+(x) = \max\{0, f(x)\} \text{ positive part of } f$$

$$f^-(x) = -\min\{0, f(x)\} \text{ negative part of } f$$

Notice that both f^+ and f^- are non-negative valued.

$$f^+, f^- : A \rightarrow [0, \infty], f = f^+ - f^-, |f| = f^+ + f^-$$

Therefore f is measurable iff f^+, f^- are measurable.

Definition 3.22. $(X, \mathcal{A}), f : A \rightarrow [-\infty, \infty], A \in \mathcal{A}$. We say f is simple iff it takes on finitely many values $\alpha_1, \alpha_2, \dots, \alpha_n$.

Remark 3.23. Let f be as before. Assume $\alpha_i \neq \alpha_j$. $A_i = \{x \in A : f(x) = \alpha_i\}$. Then

$$f = \sum_{i=1}^n \alpha_i \chi_{A_i} \text{ (standard form).}$$

Notice that f is measurable iff each $A_i \in \mathcal{A}$.

Lemma 3.24. $(X, \mathcal{A}), A \in \mathcal{A}, f : A \rightarrow \mathbb{R}$ measurable and bounded. Then for all $\varepsilon > 0$, there exists measurable simple $\varphi_\varepsilon, \psi_\varepsilon : A \rightarrow \mathbb{R}$ such that

1. $\varphi_\varepsilon \leq f \leq \psi_\varepsilon$
2. $\psi_\varepsilon - \varphi_\varepsilon < \varepsilon$

Proof. Say $f(A) \subseteq [a, b]$ and let $a = y_0 < y_1 < \dots < y_n = b$ be a partition such that $y_k - y_{k-1} < \varepsilon$. Let $A_k = f^{-1}([y_{k-1}, y_k]) \in \mathcal{A}$, then

$$\varphi_\varepsilon := \sum_{k=1}^n y_{k-1} \chi_{A_k}, \quad \psi_\varepsilon := \sum_{k=1}^n y_k \chi_{A_k}$$

so that $\varphi_\varepsilon, \psi_\varepsilon$ are simple and measurable (by last Remark).

Let $x \in A$ so that $x \in A_k$ for some k . Then, $\varphi_\varepsilon(x) = y_{k-1} \leq f(x) < y_k = \psi_\varepsilon(x)$, and $\psi_\varepsilon(x) - \varphi_\varepsilon(x) = y_k - y_{k-1} < \varepsilon$. □

Theorem 3.25 (Simple Approximation). $(X, \mathcal{A}), A \in \mathcal{A}, f : A \rightarrow \mathbb{R}$. Then, f is measurable iff there exists a sequence $\varphi_n : A \rightarrow \mathbb{R}$ of measurable simple functions such that

$$\varphi_n \longrightarrow f \text{ pointwise.}$$

Proof. ($\Leftarrow$) Done by last Corollary.

($\Rightarrow$) Assume f is measurable. We split it into two cases:

Case 1: $f \geq 0$. Then let $A_n = \{x \in A : f(x) \leq n\} \in \mathcal{A}$. If $\mathcal{A}' = \{A \cap A_n : A \in \mathcal{A}\}$, then $(A_n, \mathcal{A}')$ is a measurable space. If $U \subseteq \mathbb{R}$ is open, then

$$(f|_{A_n})'(U) = \underbrace{f^{-1}(U)}_A \cap A_n \in \mathcal{A}'$$

$$\Rightarrow f|_{A_n} \text{ is measurable and bounded}$$

By the lemma, there exists simple measurable φ_n, ψ_n such that $\varphi_n \leq f \leq \psi_n, \psi_n - \varphi_n < \frac{1}{n}$ on A_n . For $x \in A \setminus A_n$, let $\varphi_n(x) = n$. Then, $\varphi_n \leq f$ on A . Let $x \in A$ and $\varepsilon > 0$ be fixed. Then, $x \in A_N$ and we may assume $\frac{1}{N} < \varepsilon$. Therefore

$$0 \leq f(x) - \varphi_N(x) \leq \psi_N(x) - \varphi_N(x) < \frac{1}{N} < \varepsilon$$

$$\Rightarrow \varphi_n \rightarrow f \text{ pointwise}$$

Notice that:

(a) $\varphi_n \leq f$

(b) By splitting $\tilde{\varphi}_n = \max\{\varphi_1, \dots, \varphi_n\}$, $\varphi_n \rightarrow f$ and $(\tilde{\varphi}_n)$ increasing.

Case 2: General. We let

$$P = \{x \in A : f(x) \geq 0\}, N = \{x \in A : f(x) < 0\}$$

Consider $g = f\chi_P, h = -f\chi_N$. By Case 1, there exists sequence $(\varphi_n), (\psi_n)$ of simple measurable functions such that

$$\varphi \nearrow g, \psi_n \nearrow h \text{ pointwise}$$

$$\Rightarrow \varphi_n - \psi_n \rightarrow g - h = f \text{ pointwise}$$

Notice that

(a)

$$\begin{aligned} |\varphi_n - \psi_n| &\leq |\varphi_n| + |\psi_n| \\ &= \varphi_n + \psi_n \\ &\leq g + h \\ &= |f| \end{aligned}$$

(b) We have that f is measurable iff there exists (φ_n) simple measurable such that

- i. $\varphi_n \rightarrow f$ pointwise
- ii. $|\varphi_n| \leq |f|$

□

Goal: Construct a Lebesgue measurable $A \subseteq \mathbb{R}$ which is not Borel.

Example 3.26 (Cantor Set). Define recursively

$$\begin{aligned} C_0 &= [0, 1] \\ C_1 &= [0, \frac{1}{3}] \cup [\frac{2}{3}, 1] \\ C_2 &= [0, \frac{1}{9}] \cup [\frac{2}{9}, \frac{1}{3}] \cup [\frac{2}{3}, \frac{7}{9}] \cup [\frac{8}{9}, 1] \\ &\vdots \end{aligned}$$

We define $C = \bigcap C_n$. Therefore we have

$$\lambda(C) = \lim \lambda(C_n) = 0$$

Example 3.27 (Cantor Function). $f : [0, 1] \rightarrow [0, 1]$ via

$$\begin{aligned} C_1 &: (\frac{1}{3}, \frac{2}{3}) \mapsto \frac{1}{2} \\ C_2 &: \begin{cases} (\frac{1}{9}, \frac{2}{9}) \mapsto \frac{1}{4} \\ (\frac{1}{3}, \frac{2}{3}) \mapsto \frac{2}{4} = \frac{1}{2} \\ (\frac{7}{9}, \frac{8}{9}) \mapsto \frac{3}{4} \end{cases} \\ &\vdots \end{aligned}$$

That is, we let $f(x)$ be $\frac{1}{2^n}, \frac{3}{2^n}, \dots$ on the intervals removed from C_{n-1} to construct C_n . This defines $f|_{[0,1] \setminus C}$. Let $f(0) = 0$ and for $x \in C$,

$$f(x) = \sup\{f(t) : t \in [0, 1] \setminus C, t < x\}$$

Then, f is increasing and continuous. Since $f(0) = 0$ and $f(1) = 1$ and f is continuous, $f([0, 1]) = [0, 1]$ (IVT). Let $g : [0, 1] \rightarrow [0, 1]$ via

$$g(y) = \inf\{x \in [0, 1] : f(x) = y\}$$

Let $y \in [0, 1]$ be fixed. Let $\alpha := \inf\{x \in [0, 1] : f(x) = y\}$. Take $(x_n) \subseteq [0, 1]$ such that $f(x_n) = y$ and $x_n \rightarrow g(y)$. Therefore

$$\underbrace{f(x_n)}_y \rightarrow f(g(y)) \implies y = f(g(y))$$

Thus, g is injective.

HW: $g([0, 1]) \subseteq C$, the Cantor set.

Moreover, $y_1 \leq y_2 \implies f(g(y_1)) \leq f(g(y_2)) \implies g(y_1) \leq g(y_2)$, that is, g is increasing. Hence, g is Borel measurable.

Proposition 3.28. There exists a Lebesgue measurable $B \subseteq \mathbb{R}$ which is not Borel.

Proof. Let $A \subseteq [0, 1]$ be a non-Lebesgue measurable set (exists by Vitali's Theorem). Let $B = g(A) \subseteq C$. Therefore

$$\lambda^*(B) \leq \lambda(C) = 0 \implies \lambda^*(B) = 0 \implies B \text{ is Lebesgue measurable}$$

Suppose, for contradiction, that B is Borel. Therefore $g^{-1}(B)$ is Borel. However, $g^{-1}(B) = A$ as g is injective, while the preimage of Borel set is Borel. Contradict to A is non-Lebesgue-measurable. $\square$

Remark 3.29. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda)$ is not complete.

Proof. B is not Borel while C is. $\square$

3.2 Almost Everywhere

Definition 3.30. $(X, \mathcal{A}, \mu)$. Let $P(x)$ be a property on the points $x \in X$. We say P holds almost everywhere (μ -ae OR ae) iff $\exists N \in \mathcal{A}$ such that $\mu(N) = 0$, and

$$\{x \in X : P(x) \text{ false}\} \subseteq N.$$

Remark 3.31. If μ is complete, then P holds ae iff

$$\mu(\{x : P(x) \text{ false}\}) = 0.$$

Proposition 3.32. $(X, \mathcal{A}, \mu)$, $f, g : X \rightarrow [-\infty, \infty]$, $f = g$ ae. If μ is complete and f is measurable, then g is measurable.

Proof. Let $N := \{x \in X : f(x) \neq g(x)\}$. Since μ is complete, $\mu(N) = 0$. Then

$$\begin{aligned} g^{-1}[-\infty, \alpha) &= \{x \in X : g(x) < \alpha\} \\ &= \{x \in N : g(x) < \alpha\} \cup \underbrace{\{x \in N^C : f(x) < \alpha\}}_{f^{-1}[-\infty, \alpha) \cap N^C \in \mathcal{A}} \\ &\in \mathcal{A} \end{aligned}$$

$\square$

Proposition 3.33. $(X, \mathcal{A}, \mu)$, μ complete. (f_n) measurable functions, $f_n \rightarrow f$ pointwise ae. Then f is measurable.

3.3 Integration

Definition 3.34. $(X, \mathcal{A}, \mu)$. Let $f : X \rightarrow \mathbb{R}$ be non-negative, simple, and measurable. Write

$$f = \sum_{i=1}^n a_i \chi_{A_i}, A_i \in \mathcal{A}, A_i \cap A_j = \emptyset$$

We define the integral of f by

$$\int f \, d\mu = \sum_{i=1}^n a_i \mu(A_i)$$

Remark 3.35. $\int f \, d\mu$ is well-defined. Say

$$f = \sum_{i=1}^n a_i \chi_{A_i} = \sum_{j=1}^m b_j \chi_{B_j}, A_i \cap A_j = \emptyset, B_i \cap B_j = \emptyset$$

We can assume $\bigcup A_i = \bigcup B_j$, otherwise remove A_i/B_j with $f|_{A_i} = 0/f|_{B_j} = 0$. Therefore

$$\begin{aligned} \sum_{i=1}^n a_i \mu(A_i) &= \sum_i \sum_j a_i \mu(A_i \cap B_j) \\ &= \sum_i \sum_j b_j \mu(A_i \cap B_j) \\ &= \sum_j b_j \mu(B_j) \end{aligned}$$

Convention:

If we write $f = \sum a_i \chi_{A_i}$, we assume

1. $a_i \neq a_j$ nonzero (therefore $\bigcup A_i = \bigcup B_j$)
2. $A_i = \{x \in X : f(x) = a_i\}$

which we called the standard form.

Proposition 3.36. $f = \sum_{i=1}^n a_i \chi_{A_i}, g = \sum_{j=1}^m b_j \chi_{B_j}$. Then

1. $\int \alpha f \, d\mu = \alpha \int f \, d\mu$
2. $\int f + g \, d\mu = \int f \, d\mu + \int g \, d\mu$
3. If $f \leq g$, then $\int f \, d\mu \leq \int g \, d\mu$

Proof. We may only prove part 2 since the other two follow the exactly same process. Write $f + g = \sum_{i,j} (a_i + b_j) \chi_{A_i \cap B_j}$. Then

$$\begin{aligned} \int f + g \, d\mu &= \sum_{i,j} (a_i + b_j) \mu(A_i \cap B_j) \\ &= \sum_i a_i \sum_j \mu(A_i \cap B_j) + \sum_j b_j \sum_i \mu(A_i \cap B_j) \\ &= \sum_i a_i \mu(A_i) + \sum_j b_j \mu(B_j) \\ &= \int f \, d\mu + \int g \, d\mu \end{aligned}$$

□

Proposition 3.37. $(X, \mathcal{A}, \mu), f_n, f : X \rightarrow \mathbb{R}$ simple, measurable, and non-negative. If (f_n) is increasing and $f_n \rightarrow f$ pointwise, then

$$\int f d\mu = \lim_{n \rightarrow \infty} \int f_n d\mu$$

Proof. Since $(\int f_n d\mu)$ is increasing and $\int f_n d\mu \leq \int f d\mu$. By the MCT (Monotone Convergence Theorem), we have

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \sup_n \int f_n d\mu \leq \int f d\mu$$

Let $0 < \varepsilon < 1$. Let $a_1, \dots, a_k$ be the distinct values f takes on. Let $A_i = \{x \in X : f(x) = a_i\}$. Consider $A(n, i) = \{x \in A_i : f_n(x) \geq (1 - \varepsilon)a_i\}$. Since $f_n \rightarrow f$ pointwise, $A_i = \bigcup_{n=1}^{\infty} A(n, i)$. Since f_n is increasing, the $A(n, i)$ is accending (w.r.t. n). Let $g_n := \sum_{i=1}^k (1 - \varepsilon)a_i \chi_{A(n, i)}$

$$\begin{aligned} \implies \lim_{n \rightarrow \infty} \int g_n d\mu &= \lim_{n \rightarrow \infty} \sum_{i=1}^k (1 - \varepsilon)a_i \mu(A(n, i)) \\ &= \sum_{i=1}^k (1 - \varepsilon)a_i \mu\left(\bigcup_n A(n, i)\right) && , \text{ by CM} \\ &= \sum_{i=1}^k (1 - \varepsilon)a_i \mu(A_i) \\ &= (1 - \varepsilon) \int f d\mu \end{aligned}$$

Since $g_n \leq f_n$, we have

$$\begin{aligned} \int g_n d\mu &\leq \int f_n d\mu \\ \implies (1 - \varepsilon) \int f d\mu &\leq \lim \int f_n d\mu \\ \implies \int f d\mu &\leq \lim \int f_n d\mu \end{aligned}$$

□

Definition 3.38. $(X, \mathcal{A}, \mu)$. Then we say

$$L^+(X, \mathcal{A}, \mu) = \{f : X \rightarrow [-\infty, \infty] : f \text{ measurable}, f \geq 0\}$$

$$L^+(X, \mathcal{A}, \mu, \mathbb{R}) = \{f : X \rightarrow \mathbb{R} : f \text{ measurable}, f \geq 0\}$$

Then for $f \in L^+(X, \mathcal{A}, \mu)$, we define

$$\int f d\mu = \sup\left\{\int g d\mu : 0 \leq g \leq f, g \text{ simple, measurable}\right\}$$

Proposition 3.39. $f \in L^+(X, \mathcal{A}, \mu)$. If $f_n \geq 0$ are simple and measurable with (f_n) increasing and $f_n \rightarrow f$ pointwise, then

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

Proof. As before, we have $\lim \int f_n d\mu \leq \int f d\mu$.

Let $0 \leq g \leq f$ be simple and measurable. Then $g \wedge f_n$ is simple, measurable with $g \wedge f_n \rightarrow g$ pointwise. By the previous Proposition, we have

$$\begin{aligned} \int g d\mu &= \lim \int g \wedge f_n d\mu \leq \lim \int f_n d\mu \\ &\Rightarrow \int f d\mu \leq \lim \int f_n d\mu \end{aligned}$$

by taking sup. □

Proposition 3.40. $f, g \in L^+(X, \mathcal{A}, \mu, \mathbb{R}), \alpha \geq 0$. Then

1. $\int \alpha f d\mu = \alpha \int f d\mu$
2. $\int f + g d\mu = \int f d\mu + \int g d\mu$
3. If $f \leq g$, then $\int f d\mu \leq \int g d\mu$ [sup]

Proof. We again only prove part 2. $\varphi_n \nearrow f, \psi_n \nearrow g$ (Simple Approximation Theorem), then $\varphi_n + \psi_n \nearrow f + g$. Then

$$\begin{aligned} \int f + g d\mu &= \lim \int \varphi_n + \psi_n d\mu \\ &= \int f d\mu + \int g d\mu \end{aligned}$$

□

Definition 3.41. $(X, \mathcal{A}, \mu), f : X \rightarrow [-\infty, \infty]$ measurable. We say f is integrable iff $\int f^+ d\mu < \infty$ and $\int f^- d\mu < \infty$, in which case

$$\int f d\mu = \int f^+ d\mu - \int f^- d\mu$$

Let $A \in \mathcal{A}$, then we say f is integrable over A iff

$$f \cdot \chi_A \text{ is integrable}$$

We write

$$\int_A f d\mu = \int f \cdot \chi_A d\mu$$

Remark 3.42. We have

$$\int_A 1 d\mu = \int \chi_A d\mu = \mu(A)$$

Definition 3.43. We say

1. $\mathcal{L}^1(X, \mathcal{A}, \mu) = \{f : X \rightarrow [-\infty, \infty] : f \text{ is integrable}\}$
2. $\mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R}) = \{f : X \rightarrow (-\infty, \infty) : f \text{ is integrable}\}$

Proposition 3.44. Let f_1, f_2, g_1, g_2 be non-negative, real-valued, integrable functions on X . If

$$f_1 - f_2 = g_1 - g_2$$

then

$$\int f_1 d\mu - \int f_2 d\mu = \int g_1 d\mu - \int g_2 d\mu$$

Proof. $f_1 - f_2 = g_1 - g_2 \implies f_1 + g_2 = g_1 + f_2 \geq 0$, therefore

$$\begin{aligned} \int f_1 + g_2 d\mu &= \int g_1 + f_2 d\mu \\ \implies \int f_1 d\mu + \int g_2 d\mu &= \int f_2 d\mu + \int g_1 d\mu \\ \implies \int f_1 d\mu - \int f_2 d\mu &= \int g_1 d\mu - \int g_2 d\mu \end{aligned}$$

□

Proposition 3.45. $f, g \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R}), \alpha \in \mathbb{R}$. then

- a) $\alpha f, f + g \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$
- b) $\int \alpha f + g d\mu = \alpha \int f d\mu + \int g d\mu$
- c) $f \leq g \implies \int f d\mu \leq \int g d\mu$

Proof. a) HW: Scalar multiplication. We want to show that $f + g$ is integrable. That is, $\int (f + g)^+ d\mu, \int (f + g)^- d\mu < \infty$. Notice

$$(f + g)^+ = \max(f + g, 0) \leq \underbrace{\max(f, 0)}_{f^+} + \underbrace{\max(g, 0)}_{g^+}$$

$$\begin{aligned} \implies \int (f + g)^+ d\mu &\leq \int (f^+ + g^+) d\mu, \text{ as they are all non-negative} \\ &= \int f^+ d\mu + \int g^+ d\mu \\ &< \infty \end{aligned}$$

Similarly we have $\int (f + g)^- < \infty$ as desired.

b) HW: Scalar multiplication. We want to show that $\int f + g d\mu = \int f d\mu + \int g d\mu$. Notice that we have

$$\begin{aligned} f + g &= (f + g)^+ - (f + g)^- \\ f + g &= (f^+ - f^-) + (g^+ - g^-) = (f^+ + g^+) - (f^- + g^-) \end{aligned}$$

By previous Proposition, we have

$$\begin{aligned} \int (f + g)^+ d\mu - \int (f + g)^- d\mu &= \int f^+ + g^+ d\mu - \int f^- + g^- d\mu \\ \implies \int (f + g)^+ - (f + g)^- d\mu &= \int f^+ d\mu + \int g^+ d\mu - \int f^- d\mu - \int g^- d\mu \\ \implies \int (f + g) d\mu &= \int f d\mu + \int g d\mu \end{aligned}$$

c) Suppose $f \leq g$. Then $0 \leq g - f \in L^+(X, \mathcal{A}, \mu, \mathbb{R})$. By part b), we have

$$\begin{aligned} 0 &\leq \int g - f d\mu = \int g d\mu - \int f d\mu \\ \implies \int f d\mu &\leq \int g d\mu \end{aligned}$$

□

Remark 3.46. $\mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$ is an $\mathbb{R}$ -vector space.

Proposition 3.47. $(X, \mathcal{A}, \mu), f : X \rightarrow [-\infty, \infty]$ measurable. Then f is integrable iff $|f|$ is integrable. In this case, we have

$$\left| \int f d\mu \right| \leq \int |f| d\mu$$

Proof. Recall that $|f| = f^+ + f^-$.

($\implies$) Suppose that f is integrable. Then $\int f^+ d\mu, \int f^- d\mu < \infty$. Since $|f| = f^+ + f^-$ (sum of integrable functions), we conclude that $|f|$ is integrable.

($\impliedby$) Suppose $|f|$ is integrable. Then $\int |f| d\mu = \int f^+ d\mu + \int f^- d\mu < \infty$. Therefore

$$\int f^+ d\mu, \int f^- d\mu < \infty \implies f \text{ is integrable.}$$

□

Proposition 3.48 (Comparison test). $(X, \mathcal{A}, \mu)$. Let $f, g : X \rightarrow [-\infty, \infty]$ be measurable such that $|f| \leq g$. If g is integrable, then f is integrable.

Example 3.49. $f : [a, b] \rightarrow \mathbb{R}$ bounded, $|f| \leq M < \infty$. Then f is Lebesgue integrable, by taking $g(x) = M$ constant.

Proposition 3.50. Let $f \in \mathcal{L}^1(X, \mathcal{A}, \mu), g : X \rightarrow [-\infty, \infty]$ measurable. If $f = g$ ae, then $g \in \mathcal{L}^1(X, \mathcal{A}, \mu)$, and $\int f d\mu = \int g d\mu$.

Proof. 1. $f, g \geq 0$. Let $A = \{x \in X : f(x) \neq g(x)\} = X \setminus (f - g)^{-1}\{0\} \in \mathcal{A}$, and $\mu(A) = 0$. Consider $h : X \rightarrow [-\infty, \infty]$ via

$$h(x) = \begin{cases} \infty & x \in A \\ 0 & x \notin A \end{cases}$$

Let $h_n = n\chi_A$, then $h_n \rightarrow h$ pointwise, and (h_n) increasing. From before,

$$\begin{aligned} \underbrace{\int h_n d\mu}_{n\mu(A)} &\rightarrow \int h d\mu \\ \Rightarrow \int h d\mu &= 0 \end{aligned}$$

Since $f \leq g + h$, then we conclude

$$\int f d\mu \leq \int g d\mu + \underbrace{\int h d\mu}_0 = \int g d\mu$$

By symmetry, the other direction follows.

2. General case. Then $f^+ = g^+$ ae, $f^- = g^-$ ae. □

Proposition 3.51. $(X, \mathcal{A}, \mu)$, $f : X \rightarrow [-\infty, \infty]$ measurable. For $t > 0$ and $A_t = \{x \in X : f(x) \geq t\} \in \mathcal{A}$, then

$$\mu(A_t) \leq \frac{1}{t} \int_{A_t} f d\mu \leq \frac{1}{t} \int |f| d\mu$$

Proof. $t\chi_{A_t} \leq f\chi_{A_t} \leq f$ □

Corollary 3.52. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. Then $\{x \in X : f(x) \neq 0\}$ is σ -finite.

Proof. $A_n = \{x \in X : |f(x)| \geq \frac{1}{n}\}$. Then

$$\mu(A_n) \leq n \int |f| d\mu < \infty$$

Also, $\{x \in X : f(x) \neq 0\} = \bigcup A_n$. □

Corollary 3.53. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. If $\int |f| d\mu = 0$, then $f = 0$ ae. (Notice the inverse also holds by previous Proposition).

Proof. Notice that

$$\mu(\{x : |f(x)| \geq \frac{1}{n}\}) \leq n \int |f| d\mu = 0$$

By the proof in previous Corollary, $\{x \in X : f(x) \neq 0\}$ is the union. Then

$$\mu(\{x : |f(x)| \neq 0\}) = \lim_{n \rightarrow \infty} \mu(\{x : |f(x)| \geq \frac{1}{n}\}) = \lim_{n \rightarrow \infty} 0 = 0$$

□

Corollary 3.54. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. If $\int_A f d\mu \geq 0$, then $f \geq 0$ ae.

Proof. $A = \{x \in X : f(x) < 0\}$. Then

$$\begin{aligned} \int_A f d\mu &= \int f \chi_A d\mu = 0 \\ \implies \int -f \chi_A d\mu &= 0 \implies \int f \chi_A d\mu = 0 \text{ ae} \implies f \geq 0 \text{ ae} \end{aligned}$$

□

Corollary 3.55. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$. Then $|f| < \infty$ ae.

Proof. $\mu(\{x : |f(x)| \geq n\}) \leq \frac{1}{n} \int |f| d\mu$. Then

$$\mu(\{x : |f(x)| = \infty\}) \leq \frac{1}{n} \underbrace{\int |f| d\mu}_{< \infty} \longrightarrow 0$$

□

Corollary 3.56. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$, then there exists $g \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$ such that $f = g$ ae.

3.4 Limit Theorems

Theorem 3.57 (Monotone Convergence). $f, f_n \in L^+(X, \mathcal{A}, \mu)$. If (f_n) is increasing, and $f_n \rightarrow f$ pointwise. Then

$$\int f_n d\mu \longrightarrow \int f d\mu$$

Proof. We have

$$\lim \int f_n d\mu = \sup \int f_n d\mu \leq \int f d\mu$$

For each n , $f_n = \lim_{k \rightarrow \infty} g_{n,k}$, where $(g_{n,k})_k$ is an increasing sequence of non-negative, measurable, simple functions. Let

$$h_n := \max\{g_{1,n}, g_{2,n}, \dots, g_{n,n}\}$$

Then, (h_n) is an increasing sequence of measurable, simple, non-negative functions such that $h_n \leq f_n$. Moreover, $f = \lim h_n$. By a previous result, we have

$$\int f d\mu = \lim \int h_n d\mu \leq \lim \int f_n d\mu$$

□

Corollary 3.58. $f_k \in L^+(X, \mathcal{A}, \mu)$, then

$$\int \sum_{k=1}^{\infty} f_k d\mu = \sum_{k=1}^{\infty} \int f_k d\mu$$

Proof. Let

$$g_n := \sum_{k=1}^n f_k \in L^+$$

Then (g_n) is increasing and $g_n \rightarrow \sum_{k=1}^{\infty} f_k$ pointwise. Therefore

$$\lim_{n \rightarrow \infty} \int g_n d\mu = \int \sum_{k=1}^{\infty} f_k d\mu$$

Where the LHS is

$$\lim_{n \rightarrow \infty} \int g_n d\mu = \lim_{n \rightarrow \infty} \sum_{k=1}^{\infty} \int f_k d\mu = \sum_{k=1}^{\infty} \int f_k d\mu$$

□

Theorem 3.59 (Fatou's Lemma). If $(f_n) \subseteq L^+(X, \mathcal{A}, \mu)$, then

$$\int \liminf f_n d\mu \leq \liminf \int f_n$$

Proof. Let $g_n = \inf_{k \geq n} f_k$ so that $\lim g_n = \liminf f_n$. Since (g_n) is increasing, and $(g_n) \subseteq L^+$. Then

$$\begin{aligned} \int \liminf f_n d\mu &= \int \lim g_n d\mu \\ &= \lim_{n \rightarrow \infty} \int g_n d\mu \end{aligned} \quad , \text{ MCT}$$

For $k \geq n$, we have $g_n \leq f_k$, and so

$$\lim \int g_n d\mu \leq \liminf \int f_n d\mu$$

(Notice that $\liminf \int f_n d\mu := \lim_{n \rightarrow \infty} \inf_{k \geq n} \int f_k d\mu$).

□

Theorem 3.60 (Lebesgue Dominated Convergence Theorem, LDCT). Let $f, f_n : X \rightarrow [-\infty, \infty]$ be measurable. Assume $f_n \rightarrow f$ pointwise. Suppose there exists $g \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ such that

$$|f_n| \leq g, \text{ for all } n \in \mathbb{N}$$

Then $f_n, f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$ and

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \int f d\mu$$

Proof. 1. $g \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$. Since $|f_n| \leq g$ and $|f| \leq g$, $f_n, f \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$ by Comparison. Notice that $g - f_n, g - f, g + f, g + f_n \in L^+$. By Fatou,

$$\begin{aligned} \int f \, d\mu + \int g \, d\mu &= \int f + g \, d\mu \\ &\leq \liminf \int f_n + g \, d\mu \\ &= (\liminf \int f_n \, d\mu) + \int g \, d\mu \end{aligned}$$

Since $\int g \, d\mu < \infty$, we have

$$\int f \, d\mu \leq \liminf \int f_n$$

Similarly,

$$\begin{aligned} \int g - f \, d\mu &\leq \liminf \int g - f_n \, d\mu \\ \implies - \int f \, d\mu &\leq \liminf \int -f_n \, d\mu \\ \implies - \int f \, d\mu &\leq - \limsup \int f_n \\ \implies \int f \, d\mu &\geq \limsup \int f_n \end{aligned}$$

Therefore

$$\int f \, d\mu = \limsup \int f_n \, d\mu = \liminf \int f_n \, d\mu = \lim \int f_n \, d\mu.$$

2. General: HW. □

Exercise: In the LDCIT, $f_n \rightarrow f$ pointwise can be replaced with $f_n \rightarrow f$ pointwise ae.

Example 3.61. $f_n = \chi_{[n, \infty)}$, $f_n \rightarrow f, \mu = \lambda$. Then

$$0 = \int 0 \, d\lambda, \liminf \underbrace{\int f_n \, d\lambda}_{\infty} = \infty$$

3.5 Riemann Integration

Recall: $f : [a, b] \rightarrow \mathbb{R}$ bounded. We define

$$\mathcal{P} = \{a = a_0 < a_1 < \cdots < a_n = b\}$$

and

$$U(f, \mathcal{P}) = \sum M_i(a_i - a_{i-1}), L(f, \mathcal{P}) = \sum m_i(a_i - a_{i-1})$$

where

$$m_i = \inf_{x \in [a_{i-1}, a_i]} f(x), M_i = \sup_{x \in [a_{i-1}, a_i]} f(x)$$

Theorem 3.62. $f : [a, b] \rightarrow \mathbb{R}$ bounded. Then f is Riemann integrable iff $\forall \varepsilon > 0, \exists \mathcal{P}$ such that $U(f, \mathcal{P}) - L(f, \mathcal{P}) < \varepsilon$.

Theorem 3.63. $f : [a, b] \rightarrow \mathbb{R}$ bounded. Then

1. f is Riemann integrable iff f is continuous λ -ae.
2. If f is Riemann integrable, then f is Lebesgue integrable and the two integrals agree.

Proof. ($\implies$) : Suppose f is Riemann integrable. For every $n \in \mathbb{N}$, take $U(f, \mathcal{P}_n) - L(f, \mathcal{P}_n) < \frac{1}{n}$. WLOG, assume $\mathcal{P}_1 \subseteq \mathcal{P}_2 \subseteq \mathcal{P}_3 \subseteq \dots$ (we may assume that by using refinement on each $\mathcal{P}_n$). Define $g_n, h_n : [a, b] \rightarrow \mathbb{R}$ via

$$g_n(x) = \begin{cases} \inf\{f(x) : a_{i-1} \leq x \leq a_i\} & , x \in (a_{i-1}, a_i] \\ f(a) & , x = a \end{cases}$$

and

$$h_n(x) = \begin{cases} \sup\{f(x) : a_{i-1} \leq x \leq a_i\} & , x \in (a_{i-1}, a_i] \\ f(a) & , x = a \end{cases}$$

where $\mathcal{P}_n = \{a, a_1, \dots, a_{l_n}\}$. Then, (g_n) is an increasing sequence of simple, Borel measurable functions such that

$$g_n \leq f \text{ and } \int g_n d\lambda = L(f, \mathcal{P}_n)$$

Similarly, (h_n) is decreasing with

$$f \leq h_n \text{ and } \int h_n d\lambda = U(f, \mathcal{P}_n)$$

Since f is bounded, for all $x \in [a, b]$. Then $(g_n(x)), (h_n(x)) \subseteq \mathbb{R}$ are bounded. By the old MCT, we can let $g := \lim g_n, h := \lim h_n$ be the pointwise limit. Therefore g, h are Borel measurable. Say $|g|, |h| \leq M$ for some $M > 0$. Since $\int_{[a,b]} M d\lambda = M(b-a) < \infty$, we obtain that g, h are Lebesgue integrable by Comparison test. By LDCT,

$$\int g d\lambda = \lim \int g_n d\lambda = \lim L(f, \mathcal{P}_n)$$

and

$$\int h d\lambda = \lim \int h_n d\lambda = \lim U(f, \mathcal{P}_n)$$

$$\implies \int \underbrace{h - g}_{\geq 0} d\lambda = 0$$

$$\implies h = g \quad , \lambda - \text{ae}$$

If $g(x) = h(x)$, and $x \notin \underbrace{\bigcup \mathcal{P}_n}_{\text{countable}}$, then f is cts at x . Therefore f is cts λ -ae.

Also, $g \leq f \leq h$ and $g = h$ ae. Thus, $g = f$ ae. Therefore f is Lebesgue integrable and

$$\int f d\lambda = \int g d\lambda = \lim L(f, \mathcal{P}_n) = \int_a^b f(x) dx$$

($\Leftarrow$) : Suppose f is cts λ -ae. Let $\mathcal{P}_n$ be the partition of $[a, b]$ in 2^n equal-length subintervals. Let $(g_n), (h_n)$ be as before. If f is cts at x , then

$$f(x) = \lim g_n(x) = \lim h_n(x)$$

$\Rightarrow h_n - g_n \rightarrow 0$, pointwise ae By the LDCT, we have

$$\int h_n - g_n d\lambda \rightarrow 0$$

$$\Rightarrow \lim_{n \rightarrow \infty} (U(f, \mathcal{P}_n) - L(f, \mathcal{P}_n)) = 0$$

□

Example 3.64. $\chi_{\mathbb{Q}} : [0, 1] \rightarrow \mathbb{R}$ is Lebesgue integrable, and $\int \chi_{\mathbb{Q}} d\lambda = \lambda(\mathbb{Q}) = 0$, but it's not Riemann integrable.

4 Modes of Convergence

Definition 4.1. $(X, \mathcal{A}, \mu), f, f_n : X \rightarrow \mathbb{R}$ measurable. We say f_n converges to f in measurable iff $\forall \varepsilon > 0$,

$$\lim_{n \rightarrow \infty} \mu(\{x \in X : |f_n(x) - f(x)| > \varepsilon\}) = 0$$

Example 4.2. $(\mathbb{R}, \mathcal{B}(\mathbb{R}), \lambda), f_n = \chi_{[n, \infty)}$. Then $f_n \rightarrow 0$ pointwisely. But

$$\lim_{n \rightarrow \infty} \lambda(\{x : |f_n(x)| > \frac{1}{2}\}) = \lim_{n \rightarrow \infty} \lambda[n, \infty) = \infty$$

So $f_n \not\rightarrow 0$ in measure.

Example 4.3. $([0, 1], \mathcal{B}([0, 1]), \lambda)$. We define

$$\begin{aligned} f_1 &= \chi_{[0, 1)}, \\ f_2 &= \chi_{[0, \frac{1}{2})}, f_3 = \chi_{[\frac{1}{2}, 1)}, \\ f_4 &= \chi_{[0, \frac{1}{4})}, f_5 = \chi_{[\frac{1}{4}, \frac{1}{2})}, \dots \end{aligned}$$

Fix $\varepsilon > 0$. then

$$\lim_{n \rightarrow \infty} \lambda(\{x : |f_n(x)| > \varepsilon\}) = \lim_{n \rightarrow \infty} \frac{1}{2^n} = 0 \implies f_n \rightarrow 0, \text{ in measure}$$

Notice that $f_n \not\rightarrow 0$ pointwise.

Proposition 4.4. $(X, \mathcal{A}, \mu), f_n, f : X \rightarrow \mathbb{R}$ measurable. If μ is finite ($\mu(X) < \infty$) and $f_n \rightarrow f$ pointwise ae, then $f_n \rightarrow f$ in measure.

Proof. Let $\varepsilon > 0$ and let

$$A_n = \{x : |f_n(x) - f(x)| > \varepsilon\}$$

Consider

$$B_n = \bigcup_{k=n}^{\infty} A_k \in \mathcal{A}$$

. Then

$$B_1 \supseteq B_2 \supseteq B_3 \supseteq \dots$$

and

$$\begin{aligned} \bigcap B_n &\subseteq \{x \in X : f_n(x) \not\rightarrow f(x)\} \\ \implies \mu(\bigcap B_n) &= 0 \implies \lim_{n \rightarrow \infty} \mu(B_n) = 0, \text{ by CM} \end{aligned}$$

Since $A_n \subseteq B_n, \mu(A_n) \rightarrow 0$. □

Proposition 4.5. $(X, \mathcal{A}, \mu), f_n, f : X \rightarrow \mathbb{R}$ measurable. Then if $f_n \rightarrow f$ in measure, then there exists a subsequence f_{n_k} such that $f_{n_k} \rightarrow f$ pointwise ae.

Proof. We may choose $n_1 < n_2 < n_3 < \dots \in \mathbb{N}$ such that

$$\underbrace{\mu(\{x : |f_{n_k} - f(x)| > \frac{1}{k}\})}_{A_k} \leq \frac{1}{2^k}$$

Recall that

$$\limsup A_k = \{x : x \in \text{infinite many } A_k\}$$

If $x \notin \limsup A_k$, then there exists $N \in \mathbb{N}$ such that

$$k \geq N \implies |f_{n_k}(x) - f(x)| \leq \frac{1}{k}$$

Hence, $f_{n_k}(x) \rightarrow f(x)$. By A1, it is enough to prove that

$$\sum \mu(A_k) < \infty$$

(then, $\mu(\limsup A_k) = 0$). Also, since

$$\sum \mu(A_k) \leq \sum \frac{1}{2^k} < \infty,$$

we are done. □

Theorem 4.6 (Egoroff). $(X, \mathcal{A}, \mu), \mu(X) < \infty, f, f_n : X \rightarrow \mathbb{R}$ measurable. If $f_n \rightarrow f$ pointwise, then for all $\varepsilon > 0$, there exists $A \in \mathcal{A}$ such that

1. $f_n \rightarrow f$ uniformly on A , and
2. $\mu(X \setminus A) < \varepsilon$.

Proof. Let $\varepsilon > 0$ be given. Let $g_n := \sup_{i \geq n} |f_i - f|$. Thus, we have $g_n \rightarrow 0$ pointwise. Since $\mu(X) < \infty$, we have $g_n \rightarrow 0$ in measure. We may find $n_1 < n_2 < n_3 < \dots \in \mathbb{N}$ such that

$$\mu(\{x : g_n(x) > \frac{1}{k}\}) < \frac{\varepsilon}{2^k}$$

Let $B_k := \{x : g_{n_k}(x) \leq \frac{1}{k}\}$, and let $B = \bigcap B_k$. Therefore

$$\mu(B^c) = \mu(\bigcup B_k^c) \leq \sum \mu(B_k^c) < \frac{\varepsilon}{2^k} = \varepsilon$$

We claim that $f_{n_k} \rightarrow f$ uniformly on B .

Let $\varepsilon' > 0$ be given. Take k such that $\frac{1}{k} < \varepsilon'$. For $x \in B, n \geq n_k$, then

$$|f_n(x) - f(x)| \leq g_{n_k}(x) < \frac{1}{k} < \varepsilon$$

□

Example 4.7. $f_n : [0, \infty] \rightarrow \mathbb{R}$, $f_n = \chi_{[n-1, n]}$. Then $f_n \rightarrow 0$ pointwise. Assume $A \subseteq [0, \infty]$ is Lebesgue measurable such that $f_n \rightarrow 0$ uniformly on A . Then there exists $N \in \mathbb{N}$ such that

$$|f_n| < 1 \text{ for } n \geq N \text{ and on } A$$

$$\implies A \subseteq [0, N] \implies \lambda(\mathbb{R} \setminus A) = \infty$$

Definition 4.8. $f, f_n \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$. We say $f_n \rightarrow f$ in mean iff

$$\lim_{n \rightarrow \infty} \int |f_n - f| d\mu = 0$$

Proposition 4.9. $f, f_n \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})$. If $f_n \rightarrow f$ in mean, then $f_n \rightarrow f$ in measure.

Proof. We have

$$\mu(\{x : |f_n(x) - f(x)| > \varepsilon\}) \leq \frac{1}{\varepsilon} \int |f_n - f| d\mu.$$

□

5 Measure Decomposition

5.1 Signed Measures

Definition 5.1. $(X, \mathcal{A})$. A signed measure is a function $\nu : \mathcal{A} \rightarrow [-\infty, \infty]$ such that

1. $\nu(\emptyset) = 0$
2. If $A_n \in \mathcal{A}$ and $A_n \cap A_m = \emptyset$ for $n \neq m$, then

$$\nu\left(\bigsqcup A_n\right) = \sum \nu(A_n)$$

3. ν takes at most one of the values $\pm\infty$.

Example 5.2. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$, then

$$\nu(A) = \int_A f \, d\mu = \int f \chi_A \, d\mu$$

is a signed measure.

Proof. **HW:** $\nu(\emptyset) = \int 0 \, d\mu = 0$.

Let $A_n \in \mathcal{A}$ be pairwise disjoint. Let $A = \bigsqcup A_n$. Let $B_n = \bigsqcup_{k=1}^n A_k$ so that

$$f \chi_{B_n} \rightarrow f \chi_A \quad \text{pointwise, and}$$

$$|f \chi_{B_n}| \leq |f|, |f| \in \mathcal{L}^1(X, \mathcal{A}, \mu)$$

By the LDCT, we have

$$\begin{aligned} \int f \chi_{B_n} \, d\mu &\rightarrow \int f \chi_A \, d\mu \\ \implies \int_{B_n} f \, d\mu &\rightarrow \int_A f \, d\mu \end{aligned}$$

Now,

$$\begin{aligned} \int_{B_n} f \, d\mu &= \int f \chi_{B_n} \, d\mu = \int f(\chi_{A_1} + \chi_{A_2} + \cdots + \chi_{A_n}) \, d\mu = \sum_{i=1}^n \int f \chi_{A_i} \, d\mu = \sum_{i=1}^n \int_{A_i} f \, d\mu \\ \implies \sum_{i=1}^{\infty} \nu(A_k) &= \lim_{n \rightarrow \infty} \sum_{k=1}^n \nu(A_k) = \nu(A) \end{aligned}$$

Notice that $|\nu(A)| = \left| \int_A f \, d\mu \right| \leq \int |f| \, d\mu < \infty$. □

Example 5.3. $f \in L^+(X, \mathcal{A}, \mu, \mathbb{R})$, $g \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R}) \cap L^+(X, \mathcal{A}, \mu, \mathbb{R})$. Then $\nu : \mathcal{A} \rightarrow [-\infty, \infty]$ via

$$\nu(A) = \int_A f \, d\mu - \int_A g \, d\mu$$

[possibly $\nu(A) = \infty$].

Definition 5.4. $(X, \mathcal{A}), \nu$ signed measure. Then

1. We say $A \in \mathcal{A}$ is null iff $B \in \mathcal{A}, B \subseteq A \implies \nu(B) = 0$.
2. We say $A \in \mathcal{A}$ is positive iff $B \in \mathcal{A}, B \subseteq A \implies \nu(B) \geq 0$.
3. We say $A \in \mathcal{A}$ is negative iff $B \in \mathcal{A}, B \subseteq A \implies \nu(B) \leq 0$.

Lemma 5.5. $(X, \mathcal{A}), \nu$ signed. If $A \in \mathcal{A}$ with $0 < \nu(A) < \infty$, then there exists positive $P \subseteq A$ such that $\nu(P) > 0$.

Proof. If A is positive, take $A = P$. Assume A is not positive. Then we may find $B_1 \in \mathcal{A}$ with $B_1 \subseteq A$ such that

$$\nu(B_1) \leq \max\{-1, \frac{1}{2} \inf\{\nu(B) : B \in \mathcal{A}, B \subseteq A\}\}$$

Recursively, choose

$$B_n \subseteq A \setminus \left(\bigcup_{i=1}^{n-1} B_i \right)$$

such that $\nu(B_n) \leq \max\{-1, \frac{1}{2} \inf\{\nu(B) : B \in \mathcal{A}, B \subseteq A \setminus (\bigcup_{i=1}^{n-1} B_i)\}\}$

Case 1: there exists n such that $A \setminus (\bigcup_{i=1}^{n-1} B_i)$ is positive, then

$$\nu(A \setminus (\bigcup_{i=1}^{n-1} B_i)) = \underbrace{\nu(A)}_{>0} - \underbrace{\sum_{i=1}^{n-1} \nu(B_i)}_{<0} > 0$$

Case 2: The process continues and we constructed $B_1, B_2, B_3, \dots$. Let $P = A \setminus (\bigcup B_i)$. We claim that P is positive. Suppose, for contradiction, that there exists $B \in \mathcal{A}$ such that $B \subseteq P, \nu(B) < 0$. Note that $A = P \sqcup (\bigcup B_i)$, then we have

$$\nu(A) = \nu(P) + \sum \nu(B_i) < \infty$$

Therefore $\sum \nu(B_i) < \infty$, hence $\nu(B_i) \rightarrow 0$. Therefore there exists n such that $\nu(B) < 2\nu(B_n)$.

$$\implies 2\nu(B_n) \leq \inf\{\nu(C) : C \in \mathcal{A}, C \subseteq A \setminus (\bigcup_{i=1}^{n-1} B_k)\} \leq \nu(B) \quad , \text{ contradiction}$$

Also, if $\nu(P) = 0$, then $\nu(A) = \sum \nu(B_n) < 0$, contradiction.

□

Lemma 5.6. $(X, \mathcal{A}), \nu$ signed. If $A_n \in \mathcal{A}$ are positive, then $\bigcup A_n$ is positive.

Proof. Let $B \in \mathcal{A}$ such that $B \subseteq \bigcup A_n$. Consider $B_n = B \cap (A_n \setminus \bigcup_{k=1}^{n-1} A_k)$. Then

$$\begin{aligned} B &= \bigsqcup B_n, B_n \subseteq A_n \implies \nu(B_n) \geq 0 \\ \implies \nu(B) &= \sum \nu(B_n) \geq 0 \end{aligned}$$

Notice this proof works for negative and null. □

Theorem 5.7 (Hahn Decomposition). $(X, \mathcal{A})$, ν signed. Then there exists positive $P \in \mathcal{A}$ and negative $N \in \mathcal{A}$ such that $X = P \sqcup N$. Moreover, if $x \in P' \sqcup N'$, then $P \triangle P', N \triangle N'$ are null. Where $A \triangle B = (A \cup B) \setminus (A \cap B) = (A \setminus B) \cup (B \setminus A)$.

Proof. Assume ν does not take on the value of ∞ . Let $M = \sup\{\nu(A) : A \text{ positive}\}$. Therefore, there exists $(A_n)_{n=1}^\infty \subseteq \mathcal{A}$ such that $\nu(A_n) \rightarrow M$. Let $P = \bigcup A_n$, so

$$P \text{ is positive and } \nu(P) \leq M.$$

Since

$$\nu(P) = \nu(A_n) + \nu(P \setminus A_n) \geq \nu(A_n)$$

Taking $n \rightarrow \infty$, we obtain $\nu(P) \geq M$

$$\implies M = \nu(P) < \infty$$

Let $N = X \setminus P$. Suppose, for contradiction, that there exists $E \in \mathcal{A}$ such that $E \subseteq N$ and $\nu(E) > 0$. By the lemma 5.5, there exists a positive $A \subseteq E$ such that $\nu(A) > 0$. By lemma 5.6, $P \sqcup A$ is positive and

$$\nu(P \sqcup A) = \nu(P) + \nu(A) = M + \nu(A) > M$$

Contradict to $P = M$. Therefore N is negative.

Now say $X = P' \sqcup N'$, similarly. Then

$$P \setminus P' = N' \setminus N \text{ and } P' \setminus P = N \setminus N'$$

Notice that $P \setminus P'$ is positive and $N' \setminus N$ is negative, therefore they are null. The RHS is similar. Then

$$P \triangle P' = (P \setminus P') \cup (P' \setminus P) = (N' \setminus N) \cup (N \setminus N') = N \triangle N'$$

□

Example 5.8. $f \in \mathcal{L}^1(X, \mathcal{A}, \mu)$, $\nu : \mathcal{A} \rightarrow [-\infty, \infty]$, $\nu(A) = \int_A f d\mu$. Then we let

$$P = \{x \in X : f(x) \geq 0\} \quad N = \{x \in X : f(x) < 0\}$$

For $A \subseteq P$,

$$\nu(A) = \int_A f d\mu = \int \underbrace{f \chi_A}_{\geq 0} d\mu \geq 0$$

and for $A \subseteq N$,

$$\nu(A) = \int_A f \, d\mu = \int f \chi_A \, d\mu \leq 0$$

Therefore $X = P \sqcup N$ is a Hahn decomposition. Consider

$$\nu^+ : \mathcal{A} \rightarrow [0, \infty], \nu^+(A) = \nu(A \cap P)$$

Then, ν^+ is a measure on $\mathcal{A}$ and

$$\nu^+(A) = \int_{A \cap P} f \, d\mu = \int f \chi_A \chi_P \, d\mu = \int_A f \chi_P \, d\mu = \int_A f^+ \, d\mu$$

Also, $\nu^- : \mathcal{A} \rightarrow [0, \infty], \nu^-(A) = -\nu(A \cap N)$ is a measure with

$$\nu^-(A) = \int_{A \cap N} f \, d\mu = \int f \chi_A \chi_N \, d\mu = \int_A f \chi_N \, d\mu = \int_A f^- \, d\mu$$

Finally,

$$\nu(A) = \int_A f^+ \, d\mu - \int_A f^- \, d\mu = \nu^+(A) - \nu^-(A)$$

Definition 5.9. $(X, \mathcal{A}), \mu, \nu$ signed. We say μ and ν are mutually singular, $\mu \perp \nu$, iff $X = A \sqcup B$ where A is ν -null and B is μ -null.

Remark 5.10. Let $\mu, \nu, X = A \sqcup B$ be as in the definition. For $C \in \mathcal{A}$, $C = (C \cap A) \sqcup (C \cap B)$. Then

$$\begin{aligned} \mu(C) &= \mu(C \cap A) + \underbrace{\mu(C \cap B)}_0 = \mu(C \cap A) \\ \nu(C) &= \underbrace{\nu(C \cap A)}_0 + \nu(C \cap B) = \nu(C \cap B) \end{aligned}$$

Example 5.11. ν^+, ν^- from the previous example are mutually singular.

Theorem 5.12 (Jordan Decomposition). $(X, \mathcal{A}), \nu$ signed. There exists a unique pair ν^+, ν^- of mutually singular measures such that $\nu = \nu^+ - \nu^-$.

Proof. Let $X = P \sqcup N$ be a Hahn decomposition. Let $\nu^+(A) = \nu(A \cap P)$, $\nu^-(A) = -\nu(A \cap N)$. Then, ν^+, ν^- are measures. For $A \subseteq N$:

$$\nu^+(A) = \nu(A \cap P) = \nu(\emptyset) = 0$$

For $A \subseteq P$:

$$\nu^-(A) = -\nu(A \cap N) = -\nu(\emptyset) = 0$$

Also, we have

$$\nu(A) = \nu(A \cap P) + \nu(A \cap N) = \nu^+(A) - \nu^-(A)$$

Suppose that $\nu = \mu^+ - \mu^-$, where μ^+, μ^- are mutually singular measures. Say $X = P' \sqcup N'$ such that P' is μ^- -null and N' is μ^+ -null. For $A \subseteq P'$:

$$\nu(A) = \mu^+(A) \geq 0$$

therefore P is positive. Similarly, N' is negative. Therefore $X = P' \sqcup N'$ is another Hahn decomposition. Hence $P \Delta P' = N \Delta N'$ are null. For $A \in \mathcal{A}$,

$$\begin{aligned}\mu^+(A) &= \mu^+(A \cap P') \\ &= \nu(A \cap P') \\ &= \nu(A \cap P) \quad , \text{ by null} \\ &= \nu^+(A)\end{aligned}$$

Similarly, $\mu^- = \nu^-$. □

5.2 Absolute Continuity

Definition 5.13 (Absolute Continuous). Let $(X, \mathcal{A})$ be a measurable space and μ, ν be measures. Then ν is **absolutely continuous** with respect to μ ($\nu \ll \mu$) iff $\mu(A) = 0 \implies \nu(A) = 0$.

Example 5.14. Let $f \in L^+(X, \mathcal{A}, \mathbb{R})$. Define

$$\nu(A) = \int_A f d\mu$$

Then $\mu(A) = 0 \implies \nu(A) = 0$, so $\nu \ll \mu$.

Lemma 5.15. Let $(X, \mathcal{A})$ be a measurable space, and μ, ν be measures with $\nu(X) < \infty$. Then $\nu \ll \mu$ iff

$$\forall \varepsilon > 0, \exists \delta > 0, \forall A \in \mathcal{A}, \mu(A) < \delta \implies \nu(A) < \varepsilon$$

Proof. ($\Leftarrow$) straightforward.

($\Rightarrow$) Assume $\nu \ll \mu$ and assume there exists $\varepsilon > 0$ such that there is no suitable $\delta > 0$. Then, we can find $A_k \in \mathcal{A}$ such that

$$\mu(A_k) < \frac{1}{2^k} \text{ and } \nu(A_k) \geq \varepsilon$$

Then

$$\mu\left(\bigcup_{k=n}^{\infty} A_k\right) \leq \sum_{k=n}^{\infty} \mu(A_k) < \frac{1}{2^{n-1}} \text{ and } \nu\left(\bigcup_{k=n}^{\infty} A_k\right) \geq \nu(A_k) \geq \varepsilon$$

Let

$$A = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \limsup A_k$$

By [A1Q3\(d\)](#), $\mu(A) = 0$, but

$$\nu(A) = \lim_{n \rightarrow \infty} \underbrace{\nu\left(\bigcup_{k=n}^{\infty} A_k\right)}_{\geq \varepsilon} \geq \varepsilon$$

contradiction. □

Theorem 5.16 (Radon-Nikodym Theorem). Let $(X, \mathcal{A})$ be a measurable space, ν, μ be σ -finite measures. If $\nu \ll \mu$, then there exists $g \in L^+(X, \mathcal{A}, \mathbb{R})$ such that

$$\nu(A) = \int_A g d\mu$$

for all $A \in \mathcal{A}$ and g is unique μ -ae.

Proof. 1. $\nu(X), \mu(X) < \infty$

Let

$$\mathcal{F} = \{f \in L^+ : \int_A f d\mu \leq \nu(A) \text{ for all } A \in \mathcal{A}\}$$

Claim 1: $f_1, f_2 \in \mathcal{F} \implies f_1 \vee f_2 \in \mathcal{F}$.

Let $A \in \mathcal{A}$, and let

$$A_1 = \{x \in A : f_1(x) > f_2(x)\} \quad A_2 = \{x \in A : f_1(x) \leq f_2(x)\}$$

Then

$$\int_A (f_1 \vee f_2) d\mu = \int_{A_1} f_1 d\mu + \int_{A_2} f_2 d\mu \leq \nu(A_1) + \nu(A_2) = \nu(A)$$

Furthermore, $\mathcal{F}$ is non-empty as the constant 0 function belongs to it. Let $f_n \in \mathcal{F}$ such that

$$\lim_{n \rightarrow \infty} \int f_n d\mu = \sup \left\{ \int f d\mu : f \in \mathcal{F} \right\}$$

By replacing f_n with $f_1 \vee f_2 \vee \dots \vee f_n$, we may assume (f_n) is increasing. Let $g = \lim_{n \rightarrow \infty} f_n$. Then, the monotone convergence theorem shows for any $A \in \mathcal{A}$

$$\int_A g d\mu = \lim_{n \rightarrow \infty} \int_A f_n d\mu \leq \nu(A)$$

Thus, $g \in \mathcal{F}$ and $\int g d\mu = \sup \left\{ \int f d\mu : f \in \mathcal{F} \right\}$.

Claim 2: For all $A \in \mathcal{A}$, $\nu(A) = \int_A g d\mu$

Since $g \in \mathcal{F}$, $\nu_0(A) = \nu(A) - \int_A g d\mu$ is a measure on $\mathcal{A}$. Assume $\nu_0 \neq 0$, for otherwise we are done. Then, since μ is finite, there exists $\varepsilon > 0$ such that

$$\nu_0(X) > \varepsilon \mu(X)$$

Let $X = P \sqcup N$ be a Hahn decomposition for the signed measure $\nu_0 - \varepsilon \mu$. For $A \in \mathcal{A}$, we have $\nu_0(A \cap P) \geq \varepsilon \mu(A \cap P)$. Hence for all $A \in \mathcal{A}$, we have

$$\nu(A) = \int_A g d\mu + \nu_0(A) \geq \int_A g d\mu + \nu_0(A \cap P) \geq \int_A g d\mu + \varepsilon \mu(A \cap P) = \int_A g d\mu + \varepsilon \chi_P d\mu$$

Note also that $\mu(P) > 0$ for otherwise $\mu(P) = 0$ then $\nu(P) = 0$ by absolute continuity and so $\nu_0(P) = 0 \implies \nu_0(X) - \varepsilon\mu(X) = (\nu_0 - \varepsilon\mu)(N) \leq 0$, contradiction. Since $\int g d\mu \leq \nu(X) < \infty$, $g + \varepsilon\chi_P \in \mathcal{F}$ and $\int g + \varepsilon\chi_P d\mu > \int g d\mu$, contradiction with our construction of g . Since g is integrable, g is finite ae, so we may assume $g \in L^+(X, \mathcal{A}, \mathbb{R})$.

2. In general, we have $X = \bigsqcup B_n$ with $B_n \in \mathcal{A}$ and $\mu(B_n) < \infty$ and $\nu(B_n) < \infty$. By case 1, there exists nonnegative real-valued measurable $g_n : B_n \rightarrow \mathbb{R}$ such that $\nu(A) = \int_A g_n d\mu$ for all $\mathcal{A}$ -measurable $A \subseteq B_n$. Let $g : X \rightarrow \mathbb{R}$ be such that $g|_{B_n} = g_n$, i.e. $g = \sum_{n=1}^{\infty} g_n \chi_{B_n}$ (it is a pointwise limit of measurable functions so measurable). For $A \in \mathcal{A}$,

$$\nu(A) = \sum_{n \geq 1} \nu(A \cap B_n) = \sum_{n \geq 1} \int_{A \cap B_n} g_n d\mu = \int_A g d\mu$$

For uniqueness, suppose $g, h \in L^+(X, \mathcal{A}, \mathbb{R})$ such that

$$\forall A \in \mathcal{A}, \nu(A) = \int_A g d\mu = \int_A h d\mu$$

1. $\nu(X) < \infty$

We have $\int_A g - h d\mu = 0$ for all $A \in \mathcal{A}$. By taking

$$A_1 = \{x \in X : g(x) \geq h(x)\} \quad A_2 = \{x \in X : g(x) \leq h(x)\}$$

we have

$$\int_{A_1} g - h d\mu = \int (g - h)^+ d\mu = 0 \quad \int_{A_2} g - h d\mu = \int (g - h)^- d\mu = 0$$

So, $(g - h)^+ = 0, (g - h)^- = 0$ μ -ae and $g - h = 0$ μ -ae.

2. HW

□

Remark 5.17. If we take ν to be the probability measure, then Radon-Nikodym shows we can compute probability using a integral (probability density function).

Example 5.18. Let $X = [0, 1], \mathcal{A} = \mathcal{B}([0, 1]), \lambda = \text{Lebesgue measure}, \mu = \text{counting measure}$. $\lambda \ll \mu$, μ is not σ -finite. Suppose there exists $g \in L^+(X, \mathcal{A}, \mathbb{R})$ such that

$$\lambda(A) = \int_A g d\mu$$

Then

$$0 = \lambda(\{a\}) = \int_{\{a\}} g d\mu = g(a)\mu(\{0\}) = g(a) \implies g = 0 \implies \lambda([0, 1]) = \int_{[0, 1]} 0 d\mu = 0$$

Contradiction.

Definition 5.19 (R-N dervative). Let $(X, \mathcal{A})$ be a measurable space, μ, ν be σ -finite. If $\nu \ll \mu$, then $\nu(A) = \int_A g d\mu$. We call g the **Radon-Nikodym derivative** of ν wrt μ , denoted $g = \frac{d\nu}{d\mu}$.

Let $(X, \mathcal{A})$ be a measurable space, ν be a signed measure. By Hahn's decomposition, $X = P \sqcup N$ and Jordan Decomposition gives $\nu = \nu^+ - \nu^-$ with $\nu^+(A) = \nu(A \cap P), \nu^-(A) = -\nu(A \cap N)$.

Definition 5.20 (Variation). Let $(X, \mathcal{A})$ be a measure space, ν be a signed measure. The **variation** of ν is the measure

$$|\nu| = \nu^+ + \nu^-$$

Corollary 5.21. Let $(X, \mathcal{A})$ be a measurable space, μ be a σ -finite measure, ν be a signed measure with $|\nu| \ll \mu$. If $|\nu| \ll \mu$, then there exists $f = g - h$ with at least one $g \in L^+(X, \mathcal{A}, \mathbb{R})$ or $h \in L^+(X, \mathcal{A}, \mathbb{R})$ being integrable with respect to μ such that

$$\nu(A) = \int_A f d\mu \text{ for all } A \in \mathcal{A}$$

Proof. Claim: There exists a measurable $\varphi : X \rightarrow \mathbb{R}$ such that $|\varphi(x)| = 1$ for all $x \in X$ and $\nu(A) = \int_A \varphi d|\nu|$.

Let $X = P \sqcup N$ be the Hahn decomposition for ν . Let $\varphi = \chi_P - \chi_N$. We have

$$\begin{aligned} \int_A \varphi d|\nu| &\stackrel{\text{HW}}{=} \int_A \varphi d\nu^+ + \int_A \varphi d\nu^- \\ &= \int_{A \cap P} \varphi d\nu^+ + \int_{A \cap N} \varphi d\nu^- && (\text{since } \int_{A \cap N} \varphi d\nu^+ = \int_{A \cap P} \varphi d\nu^- = 0) \\ &= \int_{A \cap P} 1 d\nu^+ + \int_{A \cap N} -1 d\nu^- \\ &= \nu^+(A \cap P) - \nu^-(A \cap N) \end{aligned}$$

Since $|\nu| \ll \mu, \nu^+, \nu^- \ll \mu$. We have

$$\mu(A) = 0 \implies |\nu|(A) = 0 \implies \nu^+(A) = \nu^-(A) = 0 \implies \nu(A) = 0$$

By R-N, there exists $g \in L^+(X, \mathcal{A}, \mathbb{R})$ such that

$$|\nu|(A) = \int_A g d\mu$$

For $A \in \mathcal{A}$,

$$\begin{aligned}
\nu(A) &= \nu^+(A) - \nu^-(A) \\
&= \nu^+(A \cap P) + \underbrace{\nu^+(A \cap N) - \nu^-(A \cap N)}_0 - \underbrace{\nu^-(A \cap P)}_0 \\
&= \nu^+(A \cap P) + \nu^-(A \cap P) - \nu^+(A \cap N) - \nu^-(A \cap N) \\
&= |\nu|(A \cap P) - |\nu|(A \cap N) \\
&= \int_{A \cap P} g d\mu - \int_{A \cap N} g d\mu \\
&= \int_A g \chi_P - g \chi_N d\mu \\
&= \int_A \varphi g d\mu
\end{aligned}$$

□

Let $f = \varphi g = g \chi_P - g \chi_N$. Since ν cannot take on both $\infty, -\infty$,

$$\nu(P) = \int_P f d\mu = \int_P g d\mu < \infty \text{ or } \int_N g d\mu < \infty$$

so at least one of $g \chi_P, g \chi_N$ is integrable with respect to μ .

Theorem 5.22 (Lebesgue Decomposition). Let ν, μ be σ -finite measures on $(X, \mathcal{A})$. Then, $\nu = \nu_a + \nu_s$ where $\nu_a \ll \mu$ and $\nu_s \perp \mu$.

Proof. Consider $\lambda = \mu + \nu$, we see that

$$\nu, \mu \ll \lambda$$

By R-N, there exists $f, g \in L^+(X, \mathcal{A}, \mathbb{R})$ such that for all $A \in \mathcal{A}$

$$\mu(A) = \int_A f d\lambda, \nu(A) = \int_A g d\lambda$$

Let $A = f^{-1}((0, \infty)), B = f^{-1}(\{0\})$ and consider

$$\nu_a(E) = \nu(E \cap A) \quad \nu_s(E) = \nu(E \cap B)$$

Clearly, $X = A \sqcup B$.

Claim 1: $\nu_s \perp \mu$

If $C \subseteq A$ with $C \in \mathcal{A}$, then

$$\nu_s(C) = \nu(C \cap B) = \nu(\emptyset) = 0$$

If $C \subseteq B$ with $C \in \mathcal{A}$, then

$$\mu(C) = \int_C f d\lambda = \int_C 0 d\lambda = 0$$

Claim 2: $\nu_a \ll \mu$

Suppose $E \in \mathcal{A}$ with $\mu(E) = 0$. Then,

$$\int f \chi_E d\lambda = \int_E f d\lambda = \mu(E) = 0 \implies f \chi_E = 0 \text{ } \lambda\text{-ae} \implies \lambda(\{x \in E : f(x) > 0\}) = \lambda(E \cap A) = 0$$

Thus,

$$\nu_a(E) = \nu(E \cap A) = \lambda(E \cap A) - \mu(E \cap A) = \lambda(E \cap A) = 0 \implies \nu_a \ll \mu.$$

□

6 L^p -Spaces

6.1 L^p -Spaces

Definition 6.1. For $1 \leq p < \infty$, let

$$\mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{R}) = \{f : X \rightarrow \mathbb{R} \text{ measurable} : |f|^p \in \mathcal{L}^1(X, \mathcal{A}, \mu, \mathbb{R})\}$$

Similarly, we define $\mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{C})$.

Remark 6.2. Let $f, g \in \mathcal{L}^p = \mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{R})$, we have

$$|f + g|^p \leq (|f| + |g|)^p \leq (2(|f| \vee |g|))^p \leq 2^p(|f|^p + |g|^p) \in \mathcal{L}^1 \implies |f + g|^p \in \mathcal{L}^1 \implies f + g \in \mathcal{L}^p$$

Clearly, $f \in \mathcal{L}^p \implies \alpha f \in \mathcal{L}^p \implies \mathcal{L}^p$ is a vector space.

Definition 6.3. We say $f \in \mathcal{L}^\infty(X, \mathcal{A}, \mu, \mathbb{R})$ iff

$$\begin{aligned} \|f\|_\infty = \inf\{M > 0 : \mu(\{x \in X : |f(x)| > M\}) = 0\} < \infty &\iff \exists M > 0, \mu(\{x \in X : |f(x)| > M\}) = 0 \\ &\iff |f| \leq M \text{ ae (essentially bounded)} \end{aligned}$$

This is also a vector space. Similarly, we define $\mathcal{L}^\infty(X, \mathcal{A}, \mu, \mathbb{C})$.

Remark 6.4. Let $f \in \mathcal{L}^\infty$. Take a decreasing sequence $c_n > 0$ such that $c_n \rightarrow \|f\|_\infty$ and $\mu(\{x \in X : |f(x)| > c_n\}) = 0$. Let $A = \bigcup_{n \geq 1} \{x \in X : |f(x)| > c_n\}$ so that $\mu(A) = 0$. For $x \notin A$, $|f(x)| \leq c_n$ for all $n \in \mathbb{N}$, so $|f(x)| \leq \|f\|_\infty$. Conversely $|f(x)| \leq \|f\|_\infty \implies x \notin A$. Thus, $|f| \leq \|f\|_\infty$ ae, i.e.

$$\mu(\{x \in X : |f(x)| > \|f\|_\infty\}) = 0$$

so the infimum is achieved and is in fact a minimum.

Problem Let $f \in \mathcal{L}^p, 1 \leq p < \infty$. Let $\|f\|_p = (\int |f|^p d\mu)^{\frac{1}{p}}$. Then, $\|f\|_p = 0 \iff \int |f|^p d\mu = 0 \iff |f|^p = 0 \text{ ae} \iff |f| = 0 \text{ ae} \iff f = 0 \text{ ae}$. Hence, $\|\cdot\|_p$ is not a norm on $\mathcal{L}^p$. We will turn $\mathcal{L}^p$ into a normed vector space L^p by quotient an equivalence relation.

Let $1 \leq p \leq \infty, f, g \in \mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{R})$, we define an equivalence relation

$$f \sim g \iff f = g \text{ ae}$$

so that $[f] = \{g \in \mathcal{L}^p : f \sim g\}$ and define

$$L^p(X, \mathcal{A}, \mu, \mathbb{R}) = \{[f] : f \in \mathcal{L}^p(X, \mathcal{A}, \mu, \mathbb{R})\}$$

We equip L^p with the vector space structure:

1. $[f] + [g] = [f + g]$
2. $\alpha[f] = [\alpha f]$

This is well defined. Suppose $f_1 \sim f_2, g_1 \sim g_2$ in $\mathcal{L}^p$ so that $f_1 = f_2$ off $N_1, g_1 = g_2$ off N_2 with $\mu(N_1) = \mu(N_2) = 0$. We have $\mu(N_1 \cup N_2) = 0$ and off of $N_1 \cup N_2$, $f_1 + g_1 = f_2 + g_2 \implies f_1 + g_1 = f_2 + g_2$ ae, so

$$[f_1] + [g_1] = [f_1 + g_1] = [f_2 + g_2] = [f_2] + [g_2]$$

Notice that if $f \sim g$ in $\mathcal{L}^p$, then

$$\|f\|_p = \|g\|_p$$

Similarly, we define $L^p(X, \mathcal{A}, \mu, \mathbb{C})$.

Example 6.5. Let $X = \mathbb{N}, \mu(A) = |A|, \mathcal{A} = \mathcal{P}(\mathbb{N}), 1 \leq p < \infty$. Let $f, g : X \rightarrow \mathbb{R}$ and $f = g$ ae. Let $f \in L^p(X, \mathcal{A}, \mu, \mathbb{R})$, then

$$\|f\|_p^p = \int |f|^p d\mu = \int_{\mathbb{N}} |f|^p d\mu = \sum_{n=1}^{\infty} \int_{\{n\}} |f|^p d\mu = \sum_{n=1}^{\infty} |f(n)|^p$$

Let $f(n) = a_n$, then $\|f\|_p = (\sum_{n=1}^{\infty} |a_n|^p)^{\frac{1}{p}} < \infty$. Thus, $L^p = \ell^p$.

Remark 6.6. Let $f \in \mathcal{L}^\infty(X, \mathcal{A}, \mu, \mathbb{R})$, then there exists $M > 0, |f| \leq M$ ae. There exists g such that $f \sim g$ and $|g| \leq M$. Thus,

$$L^\infty = \{[f] : f : X \rightarrow \mathbb{R} \text{ bounded}\}$$

Our goal is to show that $(L^p, \|\cdot\|_p)$ is a normed vector space. As a convention, we write $f = [f] \in L^p = L^p(X, \mathcal{A}, \mu, \mathbb{R})$ when talking about L^p space.

1. If $p = 1$ and $f, g \in L^1$, then

$$|f + g| \leq |f| + |g| \implies \int |f + g| d\mu \leq \int |f| d\mu + \int |g| d\mu \implies \|f + g\|_1 \leq \|f\|_1 + \|g\|_1$$

So, $\|\cdot\|_1$ is a norm.

2. If $p = \infty$ and $f, g \in L^\infty$, then $|f| \leq \|f\|_\infty, |g| \leq \|g\|_\infty$ ae, so

$$|f + g| \leq |f| + |g| \leq \|f\|_\infty + \|g\|_\infty \text{ ae}$$

Thus, by the definition of inf,

$$\|f + g\|_\infty \leq \|f\|_\infty + \|g\|_\infty \implies \|\cdot\|_\infty \text{ is a norm}$$

Definition 6.7 (Hölder Conjugates). We say $1 < p, q < \infty$ are Hölder conjugates iff $\frac{1}{p} + \frac{1}{q} = 1$. We also say $1, \infty$ are Hölder conjugates.

Lemma 6.8 (Young's Inequality). Let $1 < p, q < \infty$ be Hölder conjugates. For $a, b > 0$,

$$ab \leq \frac{a^p}{p} + \frac{b^q}{q}$$

Proof. Consider $f(x) = \frac{1}{p}x^p + \frac{1}{q} - x$ on $(0, \infty)$. We have $f'(x) = x^{p-1} - 1 \implies f$ has a minimum of $f(1) = \frac{1}{p} + \frac{1}{q} - 1 = 0$. Take $x = \frac{a}{b^{q-1}}$, then

$$f(x) = \frac{1}{p} \frac{a^p}{b^{(q-1)p}} + \frac{1}{q} - \frac{a}{b^{q-1}} \geq 0$$

Since $\frac{1}{p} + \frac{1}{q} = 1 \implies p + q = pq \implies pq - p = q$,

$$\frac{1}{p} \frac{a^p}{b^q} + \frac{1}{q} \geq \frac{a}{b^{q-1}} \implies \frac{a^p}{p} + \frac{b^q}{q} \geq ab$$

□

Proposition 6.9 (Hölder's Inequality). Let $1 \leq p, q \leq \infty$ be Hölder conjugates. If $f \in L^p$ and $g \in L^q$, then

$$fg \in L^1 \text{ with } \|fg\|_1 \leq \|f\|_p \|g\|_q$$

Proof. 1. $p = 1, q = \infty$,

$$|fg| \leq |f| \|g\|_\infty \text{ ae}$$

By comparison, $fg \in L^1$. Moreover,

$$\|fg\|_1 = \int |fg| d\mu \leq \int |f| \|g\|_\infty d\mu = \|f\|_1 \|g\|_\infty$$

2. $1 < p, q < \infty$

(a) $\|f\|_p = 1, \|g\|_q = 1$

By Young's Inequality,

$$|fg| \leq \frac{|f|^p}{p} + \frac{|g|^q}{q} \implies fg \in L^1$$

Taking the integral on both sides, we get

$$\|fg\|_1 \leq \frac{1}{p} \|f\|_p^p + \frac{1}{q} \|g\|_q^q = \frac{1}{p} + \frac{1}{q} = 1 = \|f\|_p \|g\|_q$$

(b) Assume f, g are nonzero in L^p, L^q respectively. Consider $\frac{f}{\|f\|_p}, \frac{g}{\|g\|_q}$, by previous case, $\frac{fg}{\|f\|_p \|g\|_q} \in L^1 \implies fg \in L^1$, and

$$\left\| \frac{fg}{\|f\|_p \|g\|_q} \right\|_1 \leq \left\| \frac{f}{\|f\|_p} \right\|_p + \left\| \frac{g}{\|g\|_q} \right\|_q$$

□

Proposition 6.10 (Minkowski's inequality). For $f, g \in L^p$, $\|f + g\|_p \leq \|f\|_p + \|g\|_p$

Proof. We may assume that $1 < p < \infty$. First, note that $(|f + g|^{p-1})^q = |f + g|^p$. Thus, $|f + g|^{p-1} \in L^q$. Then,

$$\begin{aligned}
\int |f + g|^p d\mu &\leq \int (|f| + |g|)|f + g|^{p-1} d\mu \\
&= \int |f| \cdot |f + g|^{p-1} d\mu + \int |g| \cdot |f + g|^{p-1} d\mu \\
&\leq \|f\|_p \cdot \| |f + g|^{p-1} \|_q + \|g\|_p \cdot \| |f + g|^{p-1} \|_q \quad , \text{ Hölder} \\
&= (\|f\|_p + \|g\|_p) \left(\int |f + g|^p d\mu \right)^{1/q} \\
&\implies \left(\int |f + g|^p d\mu \right)^{1-\frac{1}{q}} \leq \|f\|_p + \|g\|_p.
\end{aligned}$$

□

Remark 6.11. $(L^p, \|\cdot\|_p)$ is a NVS (normed vector space).

6.2 Properties of L^p

Remark 6.12. (X, d) , (x_n) Cauchy. Then there exists (x_{n_k}) such that $d(x_{n_{k+1}}, x_{n_k}) < \frac{1}{4^k}$. If $x_{n_k} \rightarrow x$, then $x_n \rightarrow x$ as well.

Theorem 6.13. L^p spaces are complete (Banach).

Proof. HW: $p = \infty$.

Assume $p < \infty$, and let $(f_n) \subseteq L^p$ such that $\|f_{n+1} - f_n\|_p \leq \frac{1}{4^n}$. Let

$$\begin{aligned}
A_n &= \{x \in X : |f_{n+1}(x) - f_n(x)| \geq \frac{1}{2^n}\} \\
&= \{x \in X : |f_{n+1}(x) - f_n(x)|^p \geq \frac{1}{2^{np}}\}
\end{aligned}$$

Therefore,

$$\begin{aligned}
\mu(A_n) &\leq 2^{np} \int |f_{n+1} - f_n|^p d\mu \\
&= 2^{np} \|f_{n+1} - f_n\|_p^p \\
&\leq 2^{np} \cdot \frac{1}{4^{np}} \\
&= \frac{1}{2^{np}}
\end{aligned}$$

Hence, $\sum \mu(A_n) < \infty \implies \mu(\limsup A_n) = 0$. For $x \notin \limsup A_n$, there exists $N \in \mathbb{N}$ such that for $n \geq N$, $d(f_{n+1}(x), f_n(x)) < \frac{1}{2^n}$. For $m \geq n \geq N$, we have

$$|f_m(x) - f_n(x)| \leq \sum_{k=n}^{\infty} \frac{1}{2^k} \rightarrow 0 \quad (n \rightarrow \infty).$$

Therefore $(f_n(x))$ is Cauchy. Say $f_n(x) \rightarrow f(x) \in \mathbb{R}$. Thus, $f_n \rightarrow f$ pointwise ae. For $l \in \mathbb{N}$, we have

$$\begin{aligned} \|f_{n+l} - f_n\|_p &\leq \|f_{n+l} - f_{n+l-1}\|_p + \cdots + \|f_{n+1} - f_n\|_p \\ &\leq \sum_{i=n}^{\infty} \frac{1}{4^i} \end{aligned}$$

Then, $|f_{n+l} - f_n|^p \rightarrow |f_n - f|^p$ pointwise ae as $l \rightarrow \infty$. Therefore,

$$\begin{aligned} \int |f_n - f|^p d\mu &\leq \liminf_{l \rightarrow \infty} \int |f_{n+l} - f_n|^p d\mu \quad , \text{ Fatou} \\ &= \liminf_{l \rightarrow \infty} \|f_{n+l} - f_n\|_p^p \\ &\leq \sum_{i=n}^{\infty} \frac{1}{4^i} \rightarrow 0 \quad (n \rightarrow \infty). \end{aligned}$$

□

Theorem 6.14. The simple functions in L^p are dense.

Proof. We split it into two cases:

Case 1 $1 \leq p < \infty$. Let $f \in L^p$. Let g_n, h_n be increasing sequences of non-negative simple functions such that $g_n \nearrow f^+, h_n \nearrow f^-$ pointwise. Consider $f_n = g_n - h_n$. Then,

$$|f_n| \leq |g_n| + |h_n| \leq f^+ + f^- = |f| \in L^p, \text{ and } |f_n - f|^p \leq 2^p |f|^p \in L^1.$$

By the LDCT,

$$\int |f_n - f|^p d\mu \rightarrow \int 0 d\mu = 0.$$

Case 2 $p = \infty$. Let $f \in L^\infty$ and $\varepsilon > 0$ be fixed. Let $- \|f\|_\infty = a_0 < a_1 < a_2 < \cdots < a_n = \|f\|_\infty$ be a partition such that $a_i - a_{i-1} < \varepsilon$. Let $A_i = f^{-1}((a_{i-1}, a_i])$ and $f_\varepsilon = \sum_{i=1}^n a_i \chi_{A_i}$. Then

$$\|f - f_\varepsilon\|_\infty \leq \varepsilon.$$

□

Definition 6.15. We say $f : [a, b] \rightarrow \mathbb{R}$ is a step function iff there exists $a = a_0 < a_1 < \cdots < a_n = b$ such that $f|_{(a_i, a_{i+1})}$ is constant.

Proposition 6.16. $1 \leq p < \infty$. The step functions are dense in $L^p([a, b])$.

Proof. Let $A \subseteq [a, b]$ be Lebesgue integrable. Let $\varepsilon > 0$ be given. We must find a step function g such that

$$\|\chi_A - g\|_p < \varepsilon.$$

Let $A \subseteq \bigcup (a_n, b_n)$ such that

$$\sum b_n - a_n < \lambda(A) + \varepsilon < \infty$$

Take $N \in \mathbb{N}$ such that

$$\sum_{n=N+1}^{\infty} b_n - a_n < \varepsilon$$

Let

$$\begin{aligned} g &= \chi_{[a,b] \cap (\bigcup_{n=1}^N (a_n, b_n))} \quad \text{step function} \\ h &= \chi_{[a,b] \cap (\bigcup_{n=1}^{\infty} (a_n, b_n))} \end{aligned}$$

Then,

$$\begin{aligned} \|\chi_A - g\|_p &\leq \|\chi_A - h\|_p + \|h - g\|_p \\ &\leq (\lambda(\bigcup_{n=1}^{\infty} (a_n, b_n) \setminus A))^{\frac{1}{p}} + (\lambda(\bigcup_{n=N+1}^{\infty} (a_n, b_n)))^{\frac{1}{p}} \\ &\leq ((\sum b_n - a_n) - \lambda(A))^{\frac{1}{p}} + (\sum_{n=N+1}^{\infty} b_n - a_n)^{\frac{1}{p}} \\ &< \varepsilon^{\frac{1}{p}} + \varepsilon^{\frac{1}{p}}. \end{aligned}$$

□

Corollary 6.17. $1 \leq p < \infty$. Then $L^p([a_n, b_n])$ is separable.

Proof. “Rational” step functions are dense. □

Example 6.18. $L^\infty([0, 1])$ is not separable.

Suppose it is, and $\{f_1, f_2, \dots\}$ is dense in L^∞ . For all $x \in [0, 1]$, there exists $\theta(x) \in \mathbb{N}$ such that

$$\|\chi_{[0,x]} - f_{\theta(x)}\|_\infty < \frac{1}{2}$$

For $x \neq y$, $\|\chi_{[0,x]} - \chi_{[0,y]}\|_\infty = 1$. By the triangle inequality, $\theta(x) \neq \theta(y)$. Therefore $\theta : [0, 1] \rightarrow \mathbb{N}$ is injection, contradiction.

Proposition 6.19. $1 \leq p < \infty$. $C([a, b])$ is dense in $L^p([a, b])$.

Note: $f \in C([a, b]) \implies |f| \leq M$ by EVT. Therefore

$$\int |f|^p d\mu < \int M^p d\mu = (b-a)M^p \implies f \in L^p.$$

Definition 6.20. A σ -algebra $\mathcal{A}$ is countably generated iff there exists a countable $C \subseteq \mathcal{A}$ such that $A = \sigma(C)$.

Example 6.21. $\mathcal{M}_{\lambda^*}$ generated by $(a, b), (a, b) \in \mathbb{Q}$.

Proposition 6.22. $(X, \mathcal{A}, \mu)$ countably generated, σ -finite. If $1 \leq p < \infty$, then $L^p(X, \mathcal{A}, \mu, \mathbb{R})$ is separable.

Proof. Page 102-104. □

Example 6.23. $(X, \mathcal{A}, \mu)$ finite. Then

1. $1 \leq p < \infty, f \in L^\infty$. Then

$$\int |f|^p d\mu \leq \int \|f\|_\infty^p d\mu = \mu(X) \|f\|_\infty^p$$

Therefore $L^\infty \subseteq L^p$

2. $1 \leq p < r < \infty, f \in L^r$. Then

$$|f|^p \in L^{\frac{r}{p}}$$

Let s be the Hölder's conjugate of $\frac{r}{p}$. Then

$$\|f\|_p^p = \| |f|^p \cdot 1 \|_1 \leq \| |f|^p \|_{\frac{r}{p}} \cdot \|1\|_s < \infty$$

Therefore $f \in L^p$, and hence $L^r \subseteq L^p$

Example 6.24. $1 \leq p < q \leq \infty$, then $L^p \subseteq L^q$. In general, there are no inclusion relations.

6.3 Dual Spaces

Recall: V, W be NVSs. A linear $T : V \rightarrow W$ is continuous iff T is bounded. ($\|T\|_{\text{op}} = \sup\{\|T(x)\| : \|x\| = 1\} < \infty$).

Definition 6.25. V NVS. The dual space of V is

$$V^* = \{T : V \rightarrow \mathbb{R} \mid T \text{ linear and continuous}\}$$

Recall: $T \in V^*$, then

$$\|T\|_{\text{op}} = \sup\{|T(v)| : \|v\| = 1\} < \infty$$

For $v \neq 0$, then

$$T\left(\frac{v}{\|v\|}\right) \leq \|T\|_{\text{op}} \implies |T(v)| \leq \|T\|_{\text{op}} \cdot \|v\|$$

Example 6.26. $1 \leq p \leq \infty, q$ be its Hölder conjugate. Fix $g \in L^q$. Consider $T_g : L^p \rightarrow \mathbb{R}$, where

$$T_g(f) = \int fg d\mu$$

Claim: $T_g \in (L^p)^*$.

1. well-defined: If $f_1 = f_2$ in L^p , then

$$\int g f_1 d\mu = \int g f_2 d\mu \implies T_g(f_1) = T_g(f_2)$$

Notice that if $g_1 = g_2$ in L^q , then $T_{g_1} = T_{g_2}$

2. linear: Notice that

$$\begin{aligned} T_g(\alpha f_1 + f_2) &= \int (\alpha f_1 + f_2) g d\mu \\ &= \alpha \int f_1 g d\mu + \int f_2 g d\mu \\ &= \alpha T_g(f_1) + T_g(f_2). \end{aligned}$$

3. Continuity: For $f \in L^p$ with $\|f\|_p = 1$. Then

$$\begin{aligned} |T_g(f)| &= \left| \int f g d\mu \right| \\ &\leq \int |f g| d\mu \\ &= \|f g\|_1 \\ &\leq \|f\|_p \cdot \|g\|_q \\ &= \|g\|_q \end{aligned}$$

Therefore $\|T_g\|_{\text{op}} \leq \|g\|_q < \infty$

Proposition 6.27. $1 < p < \infty$, q HC (Hölder conjugate). Then, $T : L^q \rightarrow (L^p)^*$ via $T(g) = T_g$, is an isometry.

Proof. Again, T is well-defined. Now,

$$\begin{aligned} T(\alpha g_1 + g_2)(f) &= \int f(\alpha g_1 + g_2) d\mu \\ &= \alpha \int f g_1 d\mu + \int f g_2 d\mu \\ &= (\alpha T_{g_1} + T_{g_2})(f) \\ &= (\alpha T(g_1) + T(g_2))(f). \end{aligned}$$

Moreover, we have

$$\|T(g)\|_{\text{op}} = \|T_g\|_{\text{op}} \leq \|g\|_q.$$

It suffices to prove that $\|T(g)\|_{\text{op}} \geq \|g\|_q$. If $g = 0$, we are done. Assume $g \neq 0$. Let

$$f = \frac{|g|^{q-1} \text{sgn}(g)}{\|g\|_q^{q-1}}$$

where

$$\operatorname{sgn}(g)(x) = \begin{cases} 1 & , g(x) \geq 0 \\ -1 & , g(x) < 0 \end{cases}$$

Therefore $|f|^p = \frac{|g|^q}{\|g\|_q^q} \in L^1 \implies f \in L^p \implies \|f\|_p^p = \int \frac{|g|^q}{\|g\|_q^q} d\mu = 1 \implies \|f\|_p = 1$. Then,

$$\begin{aligned} |T_g(f)| &= \left| \int f g d\mu \right| \\ &= \left| \int \frac{|g|^q}{\|g\|_q^{q-1}} d\mu \right| \\ &= \frac{1}{\|g\|_q^{q-1}} \|g\|_q^q \\ &= \|g\|_q \end{aligned}$$

Therefore $\|T(g)\|_{\text{op}} = \|T_g\| \geq \|g\|_q$. □

Proposition 6.28. $p = 1, q = \infty, (X, \mathcal{A}, \mu)$ is σ -finite. Then $T : L^\infty \rightarrow (L^1)^*$, $T(g) = T_g$ is an isometry.

Proof. Again, T is well-defined, linear, and $\|T(g)\|_{\text{op}} \leq \|g\|_q$. Let $\varepsilon > 0$ be given. Then, there exists $B \in \mathcal{A}$ such that $0 < \mu(B) < \infty$ such that $B \subseteq \{x : |g(x)| > \|g\|_\infty - \varepsilon\}$ ($\mu(B) < \infty$ by the assumption of σ -finite by taking $B' = B \cap A$ where $\mu(A) < \infty$, so is B').

Let $f = \operatorname{sgn}(g)\chi_B \in L^1$. Then, $\|f\|_1 = \mu(B)$, and

$$T_g(f) = \int f g d\mu = \int |g| \chi_B d\mu \geq (\|g\|_\infty - \varepsilon) \mu(B).$$

Therefore,

$$\begin{aligned} (\|g\|_\infty - \varepsilon) \mu(B) &\leq |T_g(f)| \leq \|T_g\|_{\text{op}} \cdot \|f\|_1 = \|T_g\|_{\text{op}} \mu(B) \\ \implies \|g\|_\infty &\leq \|T(g)\|_\infty. \end{aligned}$$

□

Theorem 6.29 (Riesz). Let $1 \leq p < \infty$, q HC, and $(X, \mathcal{A}, \mu)$ be σ -finite. Then $T : L^q \rightarrow (L^p)^*$ is an isometric isomorphism

Proof. It suffices to show that T is surjective, as all isometry maps are injective. Let $F \in (L^p)^*$.

Case 1: $\mu(X) < \infty$. Consider $\nu : \mathcal{A} \rightarrow \mathbb{R}$ via $\nu(A) = F(\chi_A)$. If $A = \bigsqcup A_n$, then

$$\chi_{\bigsqcup_{i=1}^n A_n} \rightarrow \chi_A \text{ pointwise, and } |\chi_{\bigsqcup_{i=1}^n A_n}| \leq 1 \in L^p.$$

$$\text{By A7, } \chi_{\bigsqcup_{i=1}^n A_n} \rightarrow \chi_A \text{ in } L^p \implies F(\chi_{\bigsqcup_{i=1}^n A_n}) \rightarrow F(\chi_A) \implies \sum_{i=1}^n F(\chi_{A_i}) \rightarrow F(\chi_A)$$

$$\implies \nu(A) = F(\chi_A) = \sum_{i=1}^{\infty} F(A_i) = \sum_{i=1}^{\infty} \nu(A_i).$$

Notice that $\nu(\emptyset) = \nu(\emptyset) + \nu(\emptyset)$, and ν is finite (as the codomain of F is $\mathbb{R}$), therefore $\nu(\emptyset) = 0$, and hence ν is a finite signed measure.

If $A \in \mathcal{A}$, with $\mu(A) = 0$, then

$$\nu(A) = F(\chi_A) = F(0) = 0.$$

For $A \in \mathcal{A}$ with $\mu(A) = 0$,

$$\begin{aligned} |\nu|(A) &= \nu^+(A) + \nu^-(A) \\ &= \nu(A \cap P) - \nu(A \cap N) \\ &= 0 - 0 \\ &= 0 \\ \implies |\nu| &\ll \mu. \end{aligned}$$

By R-N, there exists g such that $\nu(A) = \int_A g d\mu$. Since ν is finite, $g \in L^1$.

Goal: Show $g \in L^q$ and $F(f) = \int fg d\mu = T_g(f) \iff F = T(g)$.

Let $E_n := \{x \in X : |g(x)| \leq n\}$. Then $|g\chi_{E_n}| \leq n \in L^q$. Let $F_n \in (L^q)^*$, $F_n(f) = F(f\chi_{E_n})$.

Notice that if $\|f\|_p = 1$, then

$$\begin{aligned} |F_n(f)| &= |F(f\chi_{E_n})| \leq \|F\|_{\text{op}} \|f\chi_{E_n}\|_p \leq \|F\|_{\text{op}} \|f\|_p = \|F\|_p \\ \implies \|F_n\|_p &\leq \|F\|_{\text{op}}. \end{aligned}$$

If $f = \chi_A$, then

$$\begin{aligned} F_n(f) &= F(f\chi_{E_n}) = F(\chi_{A \cap E_n}) = \nu(A \cap E_n) \\ \int fg\chi_{E_n} d\mu &= \int g\chi_{A \cap E_n} d\mu = \nu(A \cap E_n) \end{aligned}$$

Therefore, $F_n(f) = \int fg\chi_{E_n} d\mu$ for all characteristic functions f . Then $F_n(f) = \int fg\chi_{E_n} d\mu$ for all simple $f \in L^p$ (linearity). Thus (HW), $F_n(f) = \int fg\chi_{E_n} d\mu$ for all $f \in L^p$ (density, cty of F_n). Thus,

$$\|g\chi_{E_n}\|_q = \|F_n\|_{\text{op}} \leq \|F\|_{\text{op}} \quad (\star)$$

.

(a) If $q < \infty$, then $g\chi_{E_n} \nearrow g$, and so $\|g\chi_{E_n}\|_q \rightarrow \|g\|_q$ by MCT. Then

$$\|g\|_q \leq \|F\|_{\text{op}} \rightarrow g \in L^q.$$

(b) If $q = \infty$. Then

$$\begin{aligned} &\mu(\{x \in X : |g(x)| > \|F\|_{\text{op}}\}) \\ &= \lim_{n \rightarrow \infty} \mu(\{x \in F_n : |g(x)| > \|F\|_{\text{op}}\}) \quad , \text{ by CM} \\ &= 0 \quad , \text{ by } \star \end{aligned}$$

$$\implies \|g\|_{\infty} \leq \|F\|_{\text{op}} < \infty$$

Thus, $F(f\chi_{E_n}) = F_n(f) = \int fg\chi_{E_n} d\mu$. Taking $n \rightarrow \infty$, $F(f) = \int fg d\mu \implies F = T(g)$ (as $f\chi_{E_n} \rightarrow f$ in L^p).

Case 2: μ is σ -finite.

Notation: $B \in \mathcal{A}$, $(B, \mathcal{A}_B, \mu_B)$. If $f : B \rightarrow \mathbb{R}$ is measurable, we let $f' : X \rightarrow \mathbb{R}$ via

$$f'(x) = \begin{cases} f(x) & , x \in B \\ 0 & , x \notin B \end{cases}$$

$F_B \in L^p(B, \mathcal{A}_B, \mu_B, \mathbb{R})^*$, $F_B(f) = F(f')$. Let $X = \bigsqcup B_i$, $\mu(B_i) < \infty$. By Case 1, there exists $g_i \in L^q(B_i)$ such that $F_{B_i}(f) = \int f g_i d\mu$, $\|g_i\|_q \leq \|F_{B_i}\|_{\text{op}}$. Let $g \in L^q(x)$ given by $g|_{B_i} = g_i$. Then,

$$F(f) = \int f g d\mu \quad (\text{HW}).$$

□

Remark 6.30. σ -finiteness is not needed for $1 < p < \infty$.

Example 6.31. $X = \{a, b\}$, $\mathcal{A} = \mathcal{P}(X)$. Let $\mu(\{a\}) = 1$, $\mu(X) = \mu(\{b\}) = \infty$, $\mu(\emptyset) = 0$. Then μ is a measure but is not σ -finite. For $f \in L^1(X, \mathcal{A}, \mu, \mathbb{R})$, then

$$\begin{aligned} \int f d\mu &= \int_{\{a\}} f(a) d\mu + \int_{\{b\}} f(b) d\mu = f(a) \cdot 1 + f(b) \cdot \infty < \infty \\ &\implies f(b) = 0 \end{aligned}$$

Similarly, if $f(b) = 0$, and $f : X \rightarrow \mathbb{R}$ is measurable, then $f \in L^1$. Then, there exists $\varphi : L^1 \rightarrow \mathbb{R}$, $\varphi(a) = f(a)$ is an isometric isomorphism, as if $|\varphi(f)| = |f(u)|$, then

$$\|f\|_1 = \int |f| d\mu = \int_{\{a\}} |f(u)| = |f(a)|.$$

Clearly, all $f : X \rightarrow \mathbb{R}$ are in L^∞ . Consider $\psi : L^\infty \rightarrow \mathbb{R}^2$, $\psi(f) = (f(a), f(b))$. Again, ψ is linear and bijective. Moreover,

$$\|f\|_\infty = \max\{|f(a)|, |f(b)|\} = \|(f(a), f(b))\|_\infty.$$

Therefore, L^∞ and $(\mathbb{R}^2, \|\cdot\|_\infty)$ are isometrically isomorphic. Thus,

$$(L^1)^* \cong \mathbb{R}^* \cong \mathbb{R}$$

$$L^\infty \cong (\mathbb{R}^2, \|\cdot\|_\infty) \cong (\mathbb{R}^2, \|\cdot\|_2) := \mathbb{R}^2$$